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Flows & Liquidity

Rising corporate financing surpluses

- With cash flows rising faster than capex, corporate financing surpluses have been rising steeply in recent quarters, approaching historically high levels outside crisis periods.
- Given the strength of AI-related capex, it might seem surprising that overall corporate cash flows have been rising faster than capex. One explanation is that AI-related capex has been crowding out other capex, thus containing the growth of overall capex at an aggregate level.
- These corporate financing surpluses reduce the need for funding by the corporate sector at an aggregate level and allow corporates to sustain their buybacks, particularly outside the tech sector.
- Momentum signals for bond futures have moved back to more elevated shorts for 10y USTs and Bunds after today's sell-off; they remain in more extreme short territory for OATs vs. Bunds and extreme long territory for energy product futures.
- In a positive development, the bitcoin price had moved above its production cost in recent days (before today's correction) after spending 280 days below.

Cross Asset Fund Flow Monitor

Current level shows the latest percentile of weekly flows; Min is denoted by 0 and Max by 1. As of 16th Sept 26.

MF & ETF Flows	Min	Max	4 wk avg (\$bn)	2025 avg (\$bn)
All Equities			12.3	8.1
All Bonds			6.1	11.3
US Equities			5.5	3.5
US Bonds			4.0	4.3
Non-US Equities			6.8	4.6
Non-US Bonds			2.1	7.0
US HG Bonds			7.8	3.4
US HY Bonds			-0.8	0.4
US Lev. Loans *			0.4	0.0
US MMFs			1.1	4.0
EM Equities			-0.6	0.9
EM Bonds			0.63	0.50
Japan Equities			1.9	-0.1
China Equities			-0.27	-0.15
Europe				
Europe Equities			-0.6	0.9
Europe Bonds			0.2	4.3
Europe HG Bonds			-0.4	0.9
Europe HY Bonds			-0.15	0.17
Europe MMFs			2.4	3.8
Other Equities			6.25	3.00

Source: Lipper, ICI, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

* US LL historical flows are monthly averages converted to weekly for comparison. China onshore A-share ETFs have been excluded.

Global Markets Strategy

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Cross Asset Positioning Monitor

Current level shows the latest percentile, Min is denoted by 0 and Max by 1.

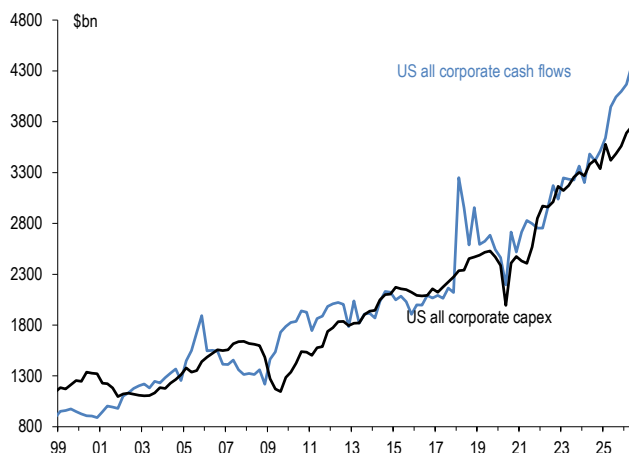
As of 22-Sep-26	MIN	MAX	Current percentile
Equities			0.77
Govt Bonds			0.62
Credit			0.30
Dollar			0.65
Commodities ex Gold			0.84
Gold			0.51
Bitcoin			0.88
EM Equities			0.77
EM Bonds/FX			0.22
Japan Equities			0.61
Europe Equities			0.72
US Equity Sectors:			
Energy			0.71
Materials			0.49
Industrials			0.35
Discretionary			0.53
Staples			0.23
Health Care			0.44
Financials			0.20
Technology			0.71
Communication Services			0.34
Utilities			0.32

Source: J.P. Morgan Flows & Liquidity.
Cross Asset Positioning Monitor aggregates across the various position indicators of Appendix ranging from positioning proxies across various futures contracts, momentum signals as proxies of how trend-following funds/CTAs are positioned, mutual fund betas as proxies of how mutual fund managers are positioned, risk parity fund positioning and leverage proxies, hedge fund betas as proxies of how hedge fund managers are positioned, client surveys, asset allocation estimates of private non-bank investors at global level, short interest indicators, etc.

- We had been arguing in previous publications that the exceptional health of the corporate sector at an aggregate level is reflected on its persistent financing surplus and on its surprisingly low interest expense. The latest Flow of Funds data for Q2 2026 reinforce this exceptional health narrative.
- As shown in Figure 1 and Figure 2, the gap between US corporate cash flows and capex, a proxy for the financing surplus, widened sharply in the first half of the year as cash flows (after taxes and dividends, and adjusting for capital consumption allowances and inventory valuation adjustments) rose much faster than capex for the US corporate sector as a whole, including domestic (or US resident) non-financials and financials. An elevated financing surplus means that the corporate sector's cash flows exceed capex and thus at an aggregate level the corporate sector is saving and in principle there is little need to raise funding at an aggregate level.

Figure 1: US all corporate capex and cash flows

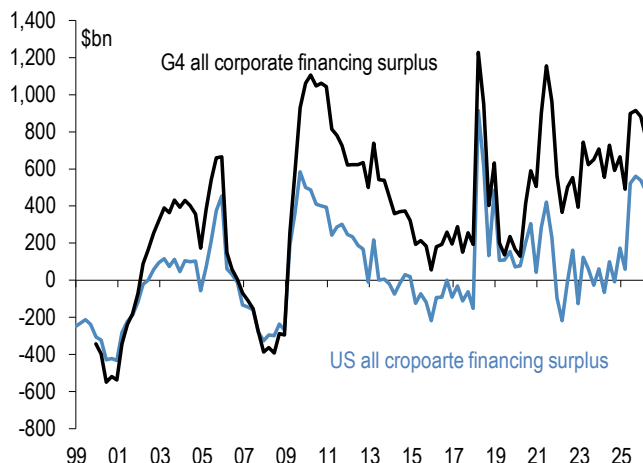
In \$bn. Last observation as of Q2 2026.



Source: Federal Reserve, J.P. Morgan Flows & Liquidity.

Figure 2: US and G4 all corporate financing surplus (i.e. cash flows minus capex)

In \$bn, US till Q2 2026, G4 till Q1 2026

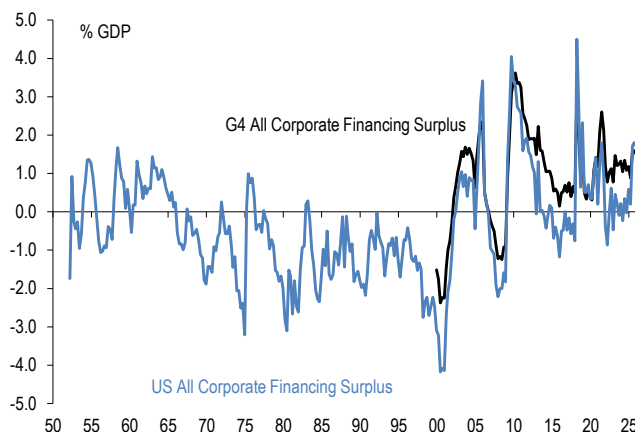


Source: Federal Reserve, J.P. Morgan Flows & Liquidity

- For G4 ex the US (Euro area, UK and Japan) corporates, we only have data till Q1 but the picture has been similar as the G4 corporate surplus has been historically dominated by US companies (Figure 2 and Figure 3). Figure 3 shows a longer history for the US corporate surplus since 1952. The Q2 surplus of close to 2% of US GDP, is close to historical highs outside crisis or repatriation periods, since data began in 1952. It was higher during crisis periods such as the Lehman crisis, but that was largely due to because of a collapse in capex. In the current juncture both capex and cash flows are rising simultaneously, but cash flows have been rising faster in recent years inducing a sharp increase in the corporate financial surplus since 2023.

Figure 3: US and G4 all corporate financing surplus (i.e. cash flows minus capex)

As % of GDP, US till Q2 2026, G4 till Q1 2026.



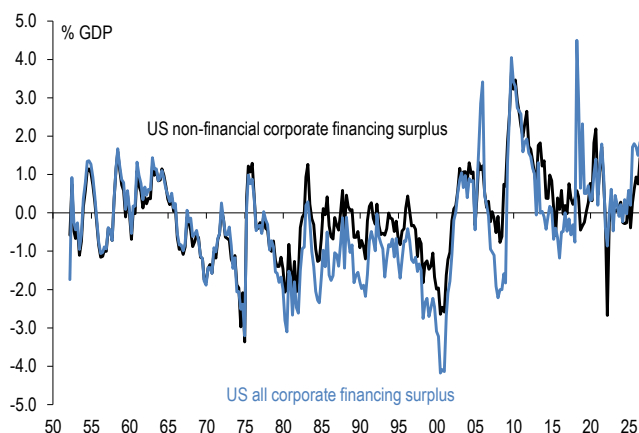
Source: Federal Reserve, J.P. Morgan Flows & Liquidity.

24 September 2026

- The financing surplus is also close to historical highs outside crisis periods even if one focuses on non-financial corporations. This is shown in Figure 4 which depicts the US surplus for both non-financial and all corporates. The difference between the two series reflects the impact of financials. The surplus of the US non-financial corporate sector stood at around 1.5% of US GDP in Q2 2026, the highest since 1958 outside crisis periods. The widening of the gap between the two series in Figure 4 in recent quarters reflects the more significant role of financials in contributing to the overall corporate surplus.

Figure 4: US all corporate and non-financial corporate financing surplus (i.e. cash flows minus capex)

As % of GDP till Q2 2026.



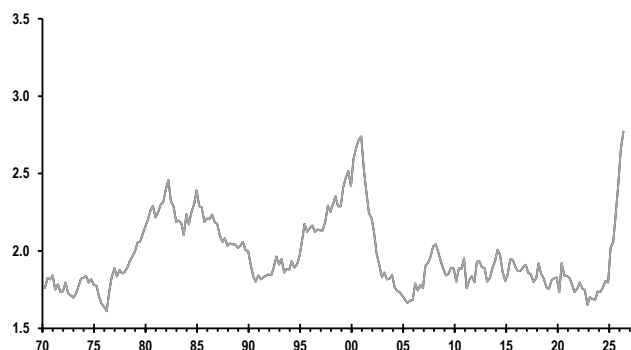
Source: Federal Reserve, J.P. Morgan Flows & Liquidity.

- Given the strength of AI-related capex, it might seem surprising that overall corporate cash flows have been rising faster than capex. One explanation is that AI-related capex has been crowding out other capex, thus containing the growth of overall capex at an aggregate level. While AI-related capex such as spending on data centres has been expanding sharply as shown in Figure 5, it has been rather concentrated and investment outside tech has been relatively softer. As a result, the overall US capex to GDP ratio has increased relatively modestly in recent years compared to the strong rise seen during the 1990s (Figure 6). This is also true if one focuses on the Intellectual Property (IP) component of capex and compares it with the steep increase seen during the 1990s as shown in Figure 7, although the slow pace of US IP in recent years may be at least partly related to the loss of bonus depreciation for R&D starting in 2022, which is brought back with OBBBA this year. This in turn has likely contributed to US IP having risen more pronouncedly in the first half of 2026. Overall, by looking at Figure 5 to Figure 7, one could argue that the capex excesses of the late 1990s, which were blamed for exacerbating the equity bubble

and the subsequent collapse at the time, are yet to be seen in the aggregate capex figures in the current juncture.

Figure 5: Computer and comm. equipment, data center, power, and comm. structures

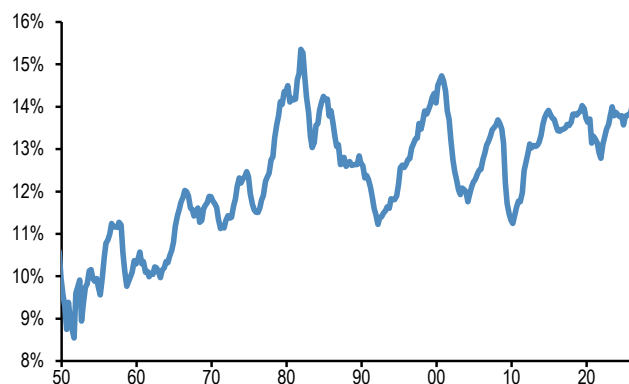
% of GDP, data centers only included from 2014.



Source: BEA, J.P. Morgan.

Figure 6: Overall US Private non-Residential Fixed Investment as % of US GDP

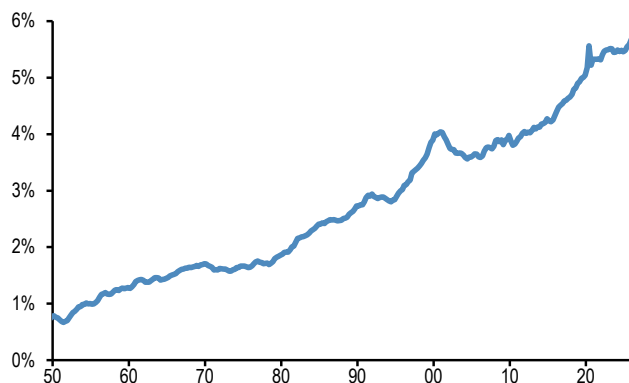
In %.



Source: BEA, J.P. Morgan.

Figure 7: Intellectual Property component of US capex as % of US GDP

In %.

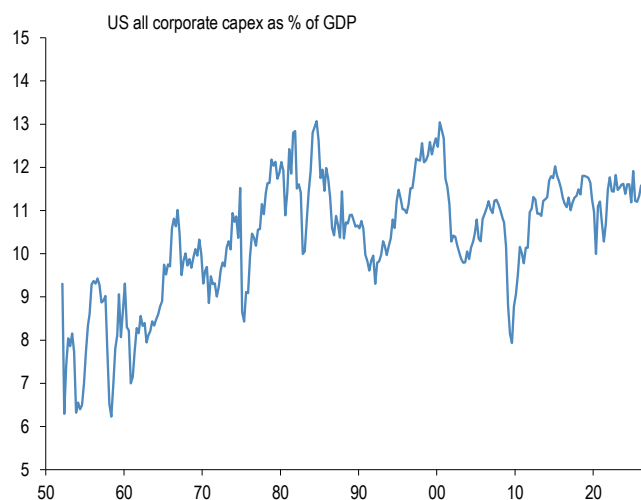


Source: BEA, J.P. Morgan.

- We get a similar picture for overall capex if we use the corporate capex component used in the financing surplus chart of Figure 3 for the US. This capex component is shown in Figure 8 which again suggests that overall US corporate capex vs GDP looks overall less exuberant than in the late 1990s.

Figure 8: US all corporate capex as % of US GDP

Till Q2 2026.

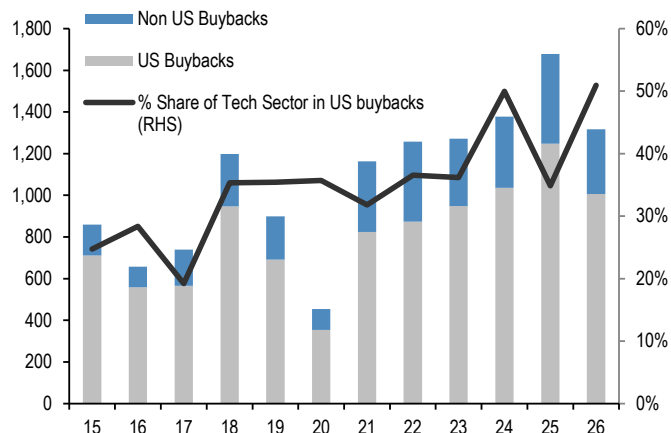


Source: Federal Reserve, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity

- Where are corporate savings deployed? They can be deployed either in equities (via share buybacks or acquisitions) thus supporting equities or in bonds via reduced debt issuance, thus supporting credit. The magnitude of US share buybacks in Figure 9 is comparable to the magnitude of corporate savings suggesting that much of US corporate savings are likely deployed into share buybacks. Indeed Figure 9 suggests that share buybacks are on track to reach \$1.7tr this year globally, \$1.3tr of which emanating from US companies.

Figure 9: Global share buyback vs Tech sector share in US buybacks

\$bn, 26 as of YTD. Buybacks are as announced.



Source: Bloomberg Finance L.P., LSEG, J.P. Morgan Flows & Liquidity.

- While we expect this corporate surplus to continue being deployed into buybacks, the sectoral composition could change going forward. As shown in Figure 9, the share of tech buybacks has been rising over the past decade. Given a likely sustained eroding of the corporate surplus of hyperscalers and other tech companies spending heavily on building AI infrastructure into 2030, and given the more significant role of financials in driving the overall US corporate surplus in recent quarters, it appears likely that the share of tech in overall buybacks could subside going forward over the next 5-10 years or so, in reversal of the past decade's rise.
- In all, with cash flows rising faster than capex, corporate financing surpluses have been rising steeply, approaching historically high levels outside crisis periods. These corporate financing surpluses reduce the need for funding by the corporate sector at an aggregate level and allow corporates to sustain their buybacks, particularly outside the tech sector.

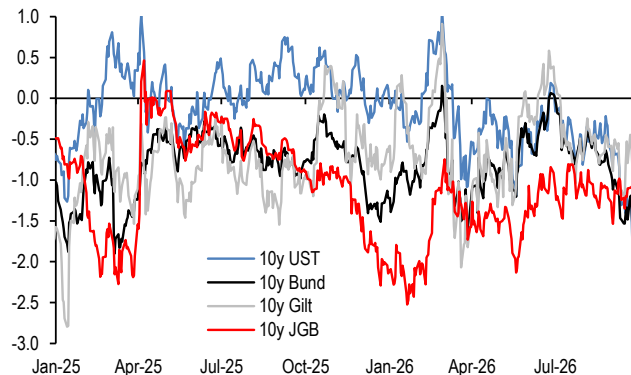
24 September 2026

Momentum signals for bond futures have moved back to more elevated shorts for 10y USTs and Bunds after today's sell-off; they remain in more extreme short territory for OATs vs. Bunds and extreme long territory for energy product futures

- Ongoing market volatility has continued to raise questions in our conversations over how positioning by momentum-based investors has evolved. We revisit below our signals outlined in Table A3 and Table A4 in the Appendix.
- As we noted last week, sharp rises in bond yields had seen momentum signals approach or reach more extreme levels where the risk of triggering mean reversion/profit taking signals becomes more elevated Figure 10 shows the average z-scores from our shorter- and longer-term momentum signals for 10y futures in USTs, Bunds, gilts and JGBs. It suggests that there had been some modest retracement in the signals for USTs and Bunds up to the close on Sep 22nd, but intraday moves after today's sharp sell-off have seen the z-scores move back towards extreme territory of -1.7 for USTs and -1.5 for Bunds. For gilt futures, the decline since last week's peak had effectively reversed the buildup of shorts since late August, though today's sell-off pushed it to somewhat more negative territory of -0.8. And for OATs vs. Bunds (Figure 11), the signals have shifted further into negative territory and sit at a more extreme level of -2.5, perhaps consistent with some further unwind of intra-EMU OW positions noted in our [European Client Survey](#) last week.

Figure 10: Momentum signals for 10y UST, Bund, gilt and JGB futures

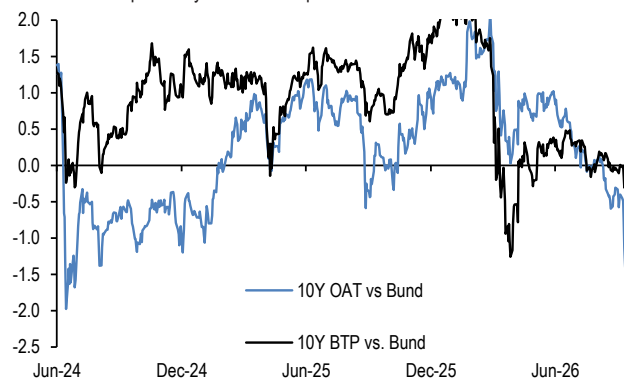
Average z-score of shorter- and longer-term momentum signal in our trend-following strategy framework (Table A3 and A4 in the Appendix), adjusted for the ratio of volatilities of the past 10 years and the past 6 months.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 11: Momentum signals for 10y OAT vs Bund futures and 10y BTP vs Bund futures

Average z-score of shorter- and longer-term momentum signal in our trend-following strategy framework (Table A3 and A4 in the Appendix), adjusted for the ratio of volatilities of the past 10 years and the past 6 months.

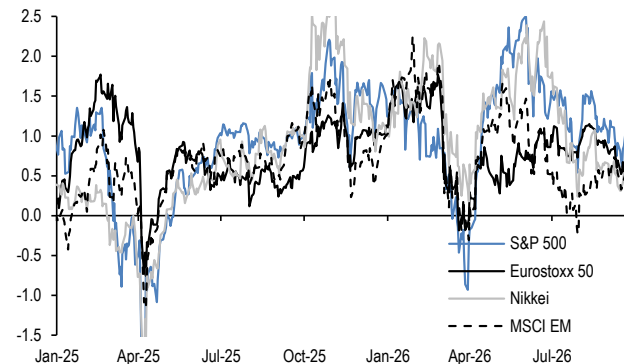


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Turning to equities, Figure 12 shows the average of the shorter- and longer-term signals futures on the S&P, Eurostoxx 50, Nikkei and MSCI EM. It suggests that after a modest decline in momentum signals since mid-August until late last week, the average z-score for the S&P rose this week from a trough of 0.6 to around 1.0 at the time of writing. In other words, the signals have shifted to somewhat elevated levels, though we are some way from more extreme levels of above 1.5 earlier in the year. For Eurostoxx 50, Nikkei and MSCI EM futures there has been relatively little change with the momentum signals clustered around 0.4-0.6.

Figure 12: Momentum signals for S&P 500, Eurostoxx 50, Nikkei and MSCI EM futures

Average z-score of shorter- and longer-term momentum signal in our trend-following strategy framework (Table A3 and A4 in the Appendix), adjusted for the ratio of volatilities of the past 10 years and the past 6 months.



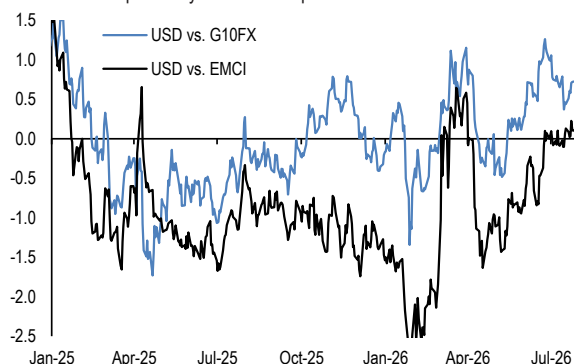
Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- In FX, Figure 13 shows the average of the shorter- and longer-term signals for the dollar vs. G10FX and vs. the EMCI. It suggests that there has been some addition of

dollar longs vs. DM FX, though the z-score is at a relatively modest level at +0.5. Against the EMCI, the signals are relatively neutral for the dollar. For the yen, the momentum signals have also increased notably from around -1.4 in late July to modest longs of around +0.5 on Sep 8th (Figure 14). This is consistent with the rise in the CFTC net spec position in the yen the following week. Since then, however, the momentum signal has shifted back to short territory and is currently at around -1.0.

Figure 13: Momentum signals for USD vs. G10FX and EMCI

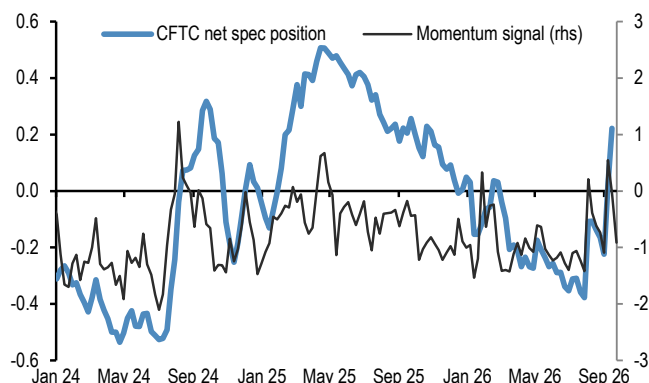
Average z-score of shorter- and longer-term momentum signal in our trend-following strategy framework (Table A3 and A4 in the Appendix), adjusted for the ratio of volatilities of the past 10 years and the past 6 months.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Figure 14: Net speculative positions and momentum signals for CME yen futures

In %, net longs as a share of open interest (lhs), and average z-score of shorter- and longer-term momentum signal in our trend-following strategy framework (Table A3 and A4 in the Appendix), adjusted for the ratio of volatilities of the past 10 years and the past 6 months (rhs).



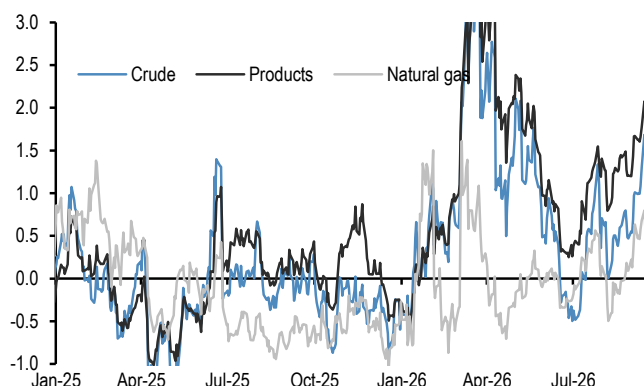
Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Turning to commodities, Figure 15 shows the average of the shorter- and longer-term signals for futures on crude (WTI and Brent), products (RBOB gasoline, ULSD and gasoil) and natural gas (NYM and TTF). It shows that for crude contracts average z-score reached a more extreme level of around +1.7 on Sep 15th, but has since reversed

some of the prior rises and are now sitting at +0.9. For products, the signals reached +2.2 on Sep 15th, and while there has been some reversal it has been more modest and remains at +1.9. For natural gas contracts, the rises have been more modest overall, though this masks a significant rise in the z-score for the TTF signal which reached a high of around +2.3 at the peak, but the rally has seen a notable reversal with the average of the shorter- and longer-term signals sitting at +0.8 at the time of writing.

Figure 15: Momentum signals for energy futures (grouped into crude, products and natural gas)

Average z-score of short- and long-term momentum signal in our trend following strategy framework, adjusted for the ratio of volatilities of the past 10 years and the past 6 months.



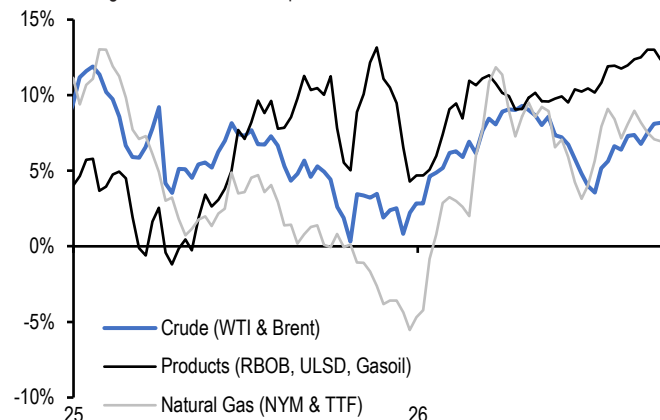
Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Looking at net spec positions from the CFTC data, the picture is similar. Figure 16 shows the ratio of the sum of the net speculative positions, proxied by the managed money category for CFTC futures complemented by the investment funds category in MiFID II disclosures for Dutch TTF futures, and the sum of total open interest in \$bns to estimate the net spec position as a % of open interest. It suggests that net longs in product contracts remain more elevated than crude and natural gas contracts.

24 September 2026

Figure 16: Net speculative position proxy for Crude, Products and Natural Gas futures

In %, net longs as a share of total open interest.

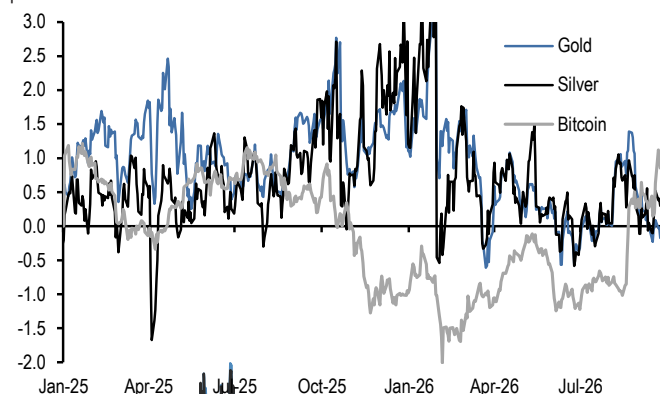


Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- Finally, Figure 17 shows the momentum signals for gold, silver and bitcoin futures. The signals for precious metals had approached more elevated levels in late August with gold reaching an average z-score of around 1.3, before retracing much of the move to close to neutral territory. Silver futures currently sit at modestly positive territory of 0.3. For bitcoin, after the recent rally the momentum signals are now at somewhat more elevated levels with the average z-score at +1.0.

Figure 17: Momentum signals gold, silver and bitcoin futures

Average z-score of short- and long-term momentum signal in our trend following strategy framework, adjusted for the ratio of volatilities of the past 10 years and the past 6 months.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

- In all, momentum signals for bond futures have moved back to more extreme shorts for 10y USTs and Bunds after today's sell-off, and remain in more extreme short territory for OATs vs. Bunds and extreme long territory for energy product futures.

In a positive development, the bitcoin price had moved above its production cost in recent days (before today's correction) after spending 280 days below

- The strength of the bitcoin rally over the past week despite the failure of the US Congress to advance the Clarity Act ([Cryptocurrency Markets](#), Sep 16th by our colleague Ken Worthington for further detail) is consistent with unwinding of bearish bitcoin positions as discussed in our previous publication ([Flows & Liquidity](#), Sep 16th). The recent rally (before today's correction) has also pushed spot bitcoin prices above our implied production cost, which has historically acted as a soft floor.
- Our implied average cost to produce one bitcoin stands currently at around \$85k, and spot prices had recently moved above that level before today's correction (Figure 18). Under this framework, the bitcoin price spent previously 280 days below the production cost pushing high cost miners to unprofitable territory. To the extent it is sustained, this new backdrop should provide relief to bitcoin miners, thus reducing the risk of forced selling by them. Operationally, miners have been managing fleets by redeploying rigs to lower-cost power regions, reselling older rigs, putting rigs on standby, and scrapping or recycling less efficient equipment. This cleanup can raise average efficiency and also help to reduce selling pressure from weaker miners.

Figure 18: Implied bitcoin cost of production

In \$



Source: Coin Metrics, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

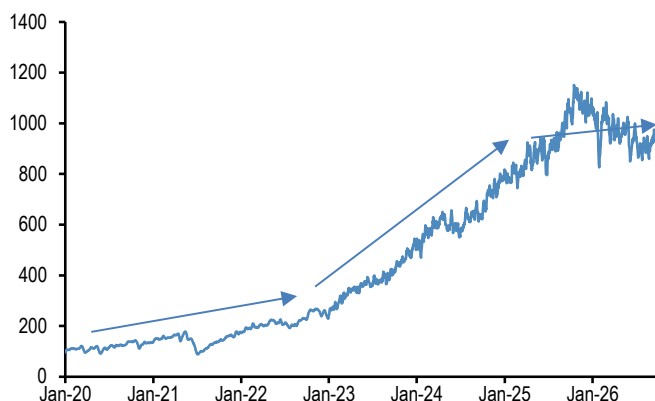
- The last time the bitcoin price spent such a long period below its production cost was in 2018. At the time bitcoin traded below estimated production cost for roughly 224 days, driven by a sharp down-cycle and the unwind of prior speculative buying. Those price declines induced high cost miners to shut down, thus reducing the hash rate and

network difficulty. While currently the mining industry is larger and more industrialized, the same stabilizing mechanism to bitcoin hash rate and mining network difficulty still applies as high cost miners retreat. With prices now rising above production cost, this mechanism that has served to put downward pressure on mining activity may be behind us for now.

- Outside this price driven -opportunistic retreat, it is worth emphasizing that the bitcoin mining industry has been also undergoing a structural shift towards AI. As miners have been pivoting either partially or fully toward AI, the trend in the bitcoin network hashrate has structurally slowed. This is reflected in the roughly 19% and 15% declines in network hashrate and mining difficulty, respectively, from their peak last October (Figure 19 and Figure 20). Consistent with this, most public miners have reduced their hashrate growth forecasts as long-term AI contracts accelerate the shift away from bitcoin mining. AI customers are paying substantial premiums for existing power access and data-centre-ready infrastructure. With bitcoin prices having been subdued for much of this year, miners have been attracted to AI cash flows that are more predictable, more stable, and higher per megawatt than mining. As a result of this structural pivot to AI, publicly listed miners are reducing their share versus privately held and sovereign miners.

Figure 19: Bitcoin network hashrate

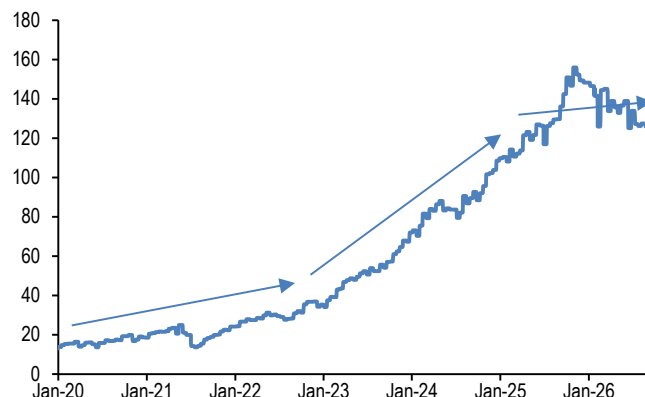
In EH/s, 7 DMA



Source: Coin Metrics, J.P. Morgan Flows & Liquidity.

Figure 20: Bitcoin network difficulty

In tr



Source: Coin Metrics, J.P. Morgan Flows & Liquidity.

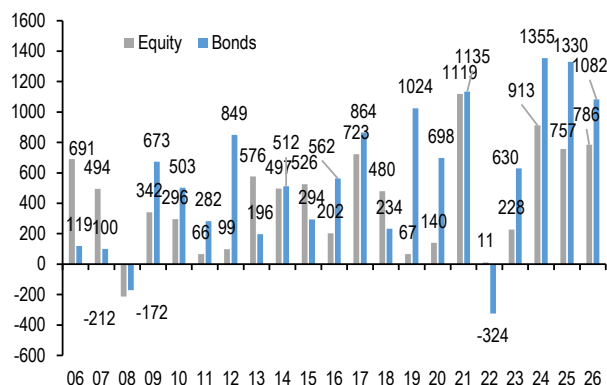
- From a bitcoin perspective, this can reduce excess hashrate growth and help prevent the network from becoming “too crowded” i.e., avoiding concentration risk. At the same time, by flattening the trend in the bitcoin hash rate, this structural shift by miners to AI implies that the bitcoin production cost would rise more slowly going forward (outside halving events).
- In all, in a positive development, the bitcoin price had moved above its production cost in recent days (before today’s correction) after spending 280 days below. To the extent it is sustained, this new backdrop should provide relief to bitcoin miners, thus reducing the risk of forced selling by them.

24 September 2026

Appendix

Chart A1a: Global equity & bond fund flows

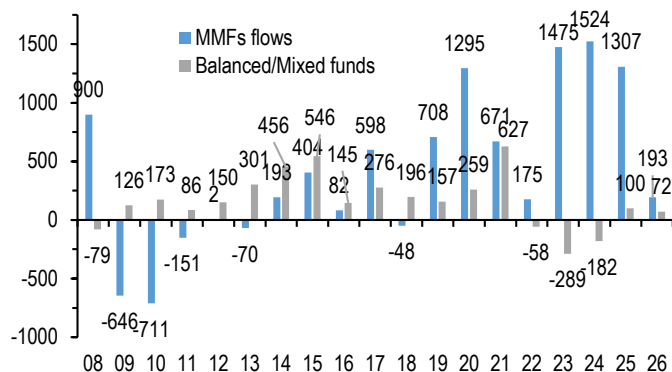
\$bn per year of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Flows come from ICI (worldwide data up to Q1'26). Data since then are a combination of monthly and weekly data from Lipper and Bloomberg.



Source: ICI, LSEG Lipper, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A1b: Quarterly Balanced/Mixed funds & MMFs flows worldwide

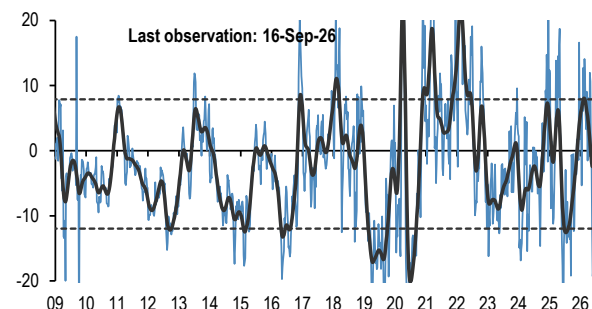
\$bn per quarter of Net Sales, i.e. includes net new sales + reinvested dividends for Mutual Funds and ETFs globally, i.e. for funds domiciled both inside and outside the US. Data come from ICI (worldwide data) and are till Q1'26.



Source: ICI, J.P. Morgan Flows & Liquidity.

Chart A2: Fund flow indicator

Difference between flows into Equity and Bond funds: \$bn per week. Difference between flows into Equity vs. Bond funds in \$bn per week. Flows include Mutual Fund and ETF flows globally, i.e. funds domiciled both inside and outside the US (source: Lipper). The thin blue line shows the 4-week average of difference between Equity and Bond fund flows. Dotted lines depict ± 1 StDev of the blue line since 2009. The thick black line shows a smoothed version of the same series. The smoothing is done using a Hodrick-Prescott filter with a Lambda parameter of 100. China onshore A-share ETFs have been excluded.



Source: LSEG Lipper, J.P. Morgan Flows & Liquidity.

Table A2: Trading turnover monitor

Volumes are monthly and Turnover ratio is annualised (monthly trading volume annualised divided by the amount outstanding). UST Cash is primary dealer transactions in all US government securities. UST futures are from Bloomberg Finance L.P. JGBs are OTC volumes in all Japanese government securities. Bunds, Gold, Oil and Copper are futures. Gold includes Gold ETFs. Min-Max chart is based on Turnover ratio. For Bunds and Commodities, futures trading volumes are used while the outstanding amount is proxied by open interest. The diamond reflects the latest turnover observation. The thin blue line marks the distance between the min and max for the complete time series since Jan-2005 onwards. Y/Y change is change in YTD notional volumes over the same period last year.

As of Aug-26	MIN	MAX	Turnover ratio	Vol (tr)	y/y chng
Equities					
EM Equity*			1.4	\$1.8	129%
DM Equity*			1.4	\$13.7	43%
Govt Bonds					
UST cash			12.2	\$17.8	8%
UST futures			0.8	\$27.1	10%
JGBs*			48.0	¥4,663	12%
Bund futures			1.3	€6.9	20%
Credit					
US HG			1.1	\$0.7	18%
US HY			0.3	\$0.1	-5%
US Convertibles			3.2	\$0.04	11%
Commodities					
Gold			54.8	\$1.9	32%
Oil			83.3	\$2.4	6%
Copper			1.9	\$0.9	68%
Digital Assets					
CME Bitcoin			120.2	\$0.084	-35%
CME Ethereum			134.8	\$0.032	-17%

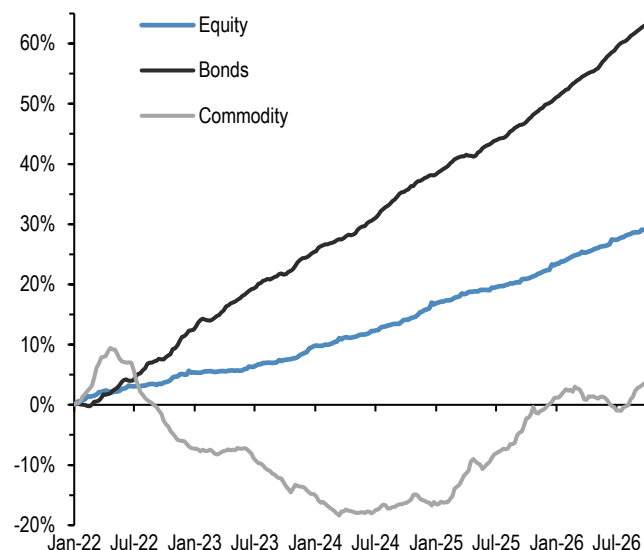
* Data with one month lag

Source: Bloomberg Finance L.P., Federal Reserve, Trace, Japan Securities Dealer Association, WFE, J.P. Morgan Flows & Liquidity.

ETF Flow Monitor (as of 23rd September)

Chart A3: Global Cross Asset ETF Flows

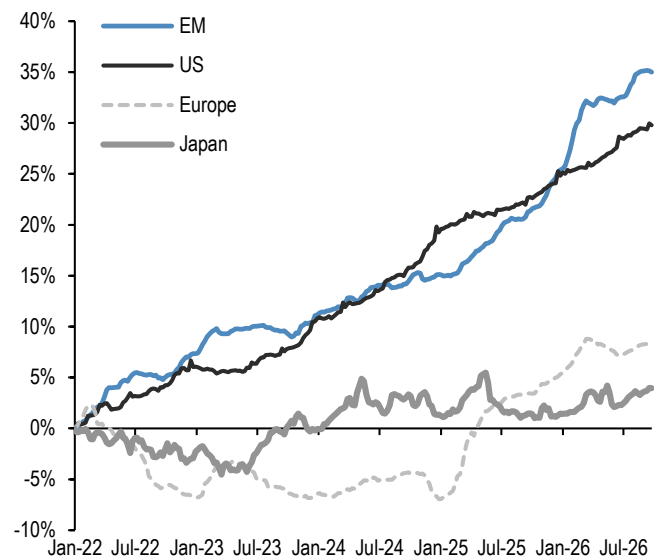
Cumulative flows into ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity. China onshore A-share ETFs have been excluded.

Chart A5: Global Equity ETF Flows

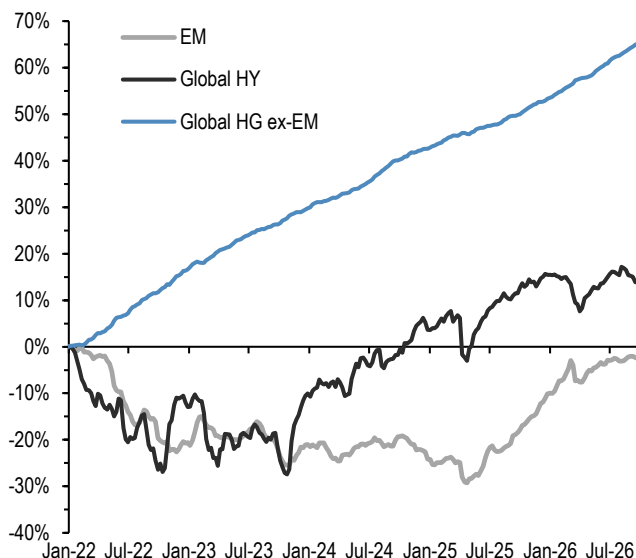
Cumulative flows into global equity ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.
 Note: We include ETFs with AUM > \$200mn in all the flow monitor charts. In Chart A5 for Japan, we subtract the BoJ buying of ETFs. China onshore A-share ETFs have been excluded.

Chart A4: Bond ETF Flows

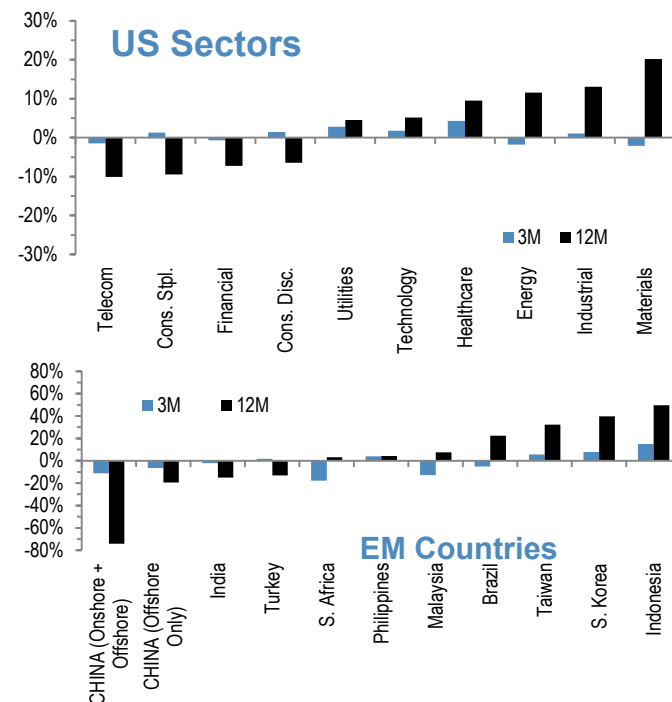
Cumulative flows into bond ETFs as a % of AUM



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A6a: Equity Sectoral and Regional ETF Flows

Rolling 3-month and 12-month change in cumulative flows as a % of AUM. Both sorted by 12-month change

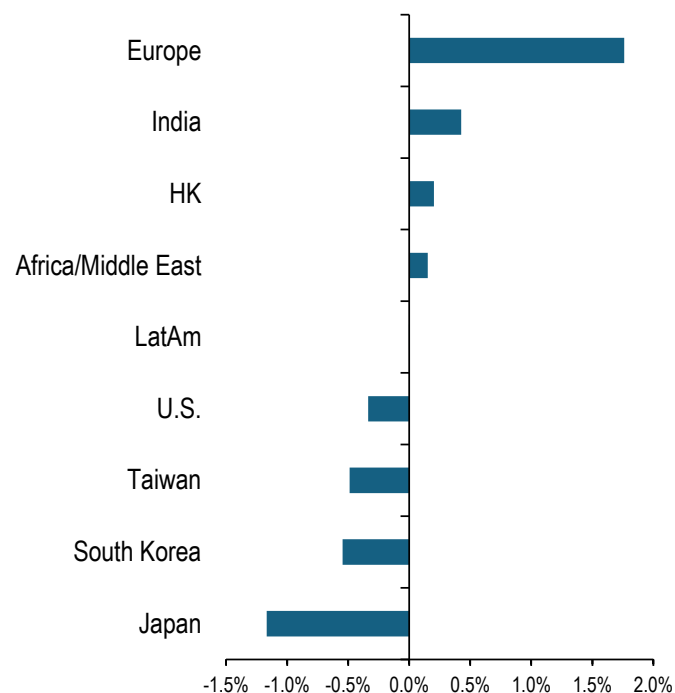


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

24 September 2026

Chart A6b: Regional Equity Allocation Monitor Implied by ETFs

In %.



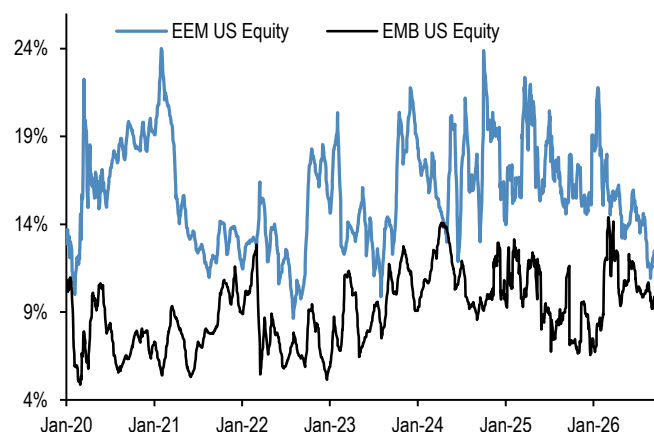
Source: Bloomberg Finance L.P., MSCI and J.P. Morgan Flows & Liquidity.

Note: Our regional equity allocation monitor implied by ETFs is based on a comparison of the share of each region in the equity ETF space relative to their corresponding shares in global equity indices.

Short Interest Monitor

Chart A7: Short interest on the EEM and EMB US ETF

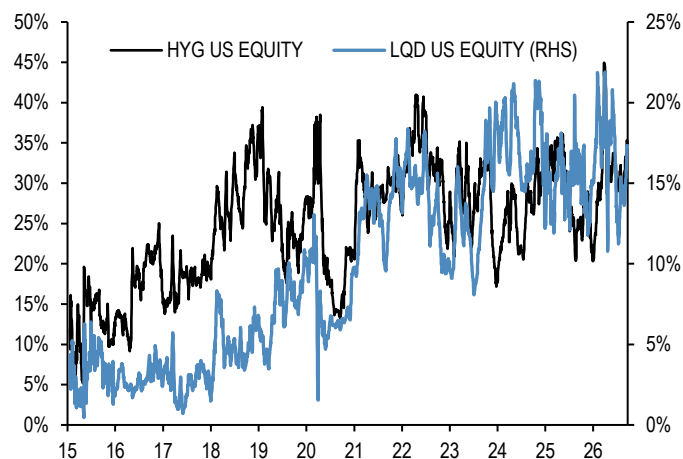
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A8: Short interest on the LQD and HYG US ETF

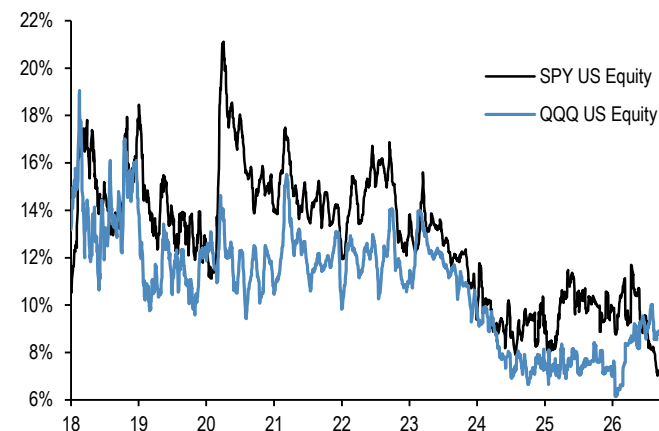
Short Interest as a % share of shares outstanding.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A9: Short interest on the SPY and QQQ US ETF

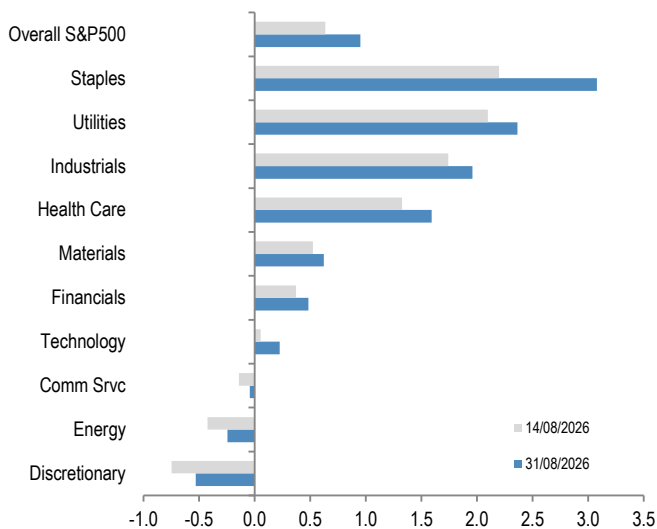
Short Interest as a % share of shares outstanding. Last obs is for 21st Sept 2026.



Source: S3, J.P. Morgan Flows & Liquidity.

Chart A10: S&P500 sector short interest

Short interest as a % of shares outstanding based on z-scores. A strategy which overweights the S&P500 sectors with the highest short interest z-score (as % of shares o/s) vs. those with the lowest, produced an information ratio of 0.7 with a success rate of 56% (see F&L, Jun 28, 2013 for more details).



Source: NYSE, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A11a: Cross Asset Volatility Monitor 3m ATM Implied Volatility (1y history), as of 21st Sep-2026

This table shows the richness/cheapness of current three-month implied volatility levels (red dot) against their one-year historical range (thin blue bar) and the ratio to current realised volatility. Assets with implied volatility outside their 25th/75th percentile range (thick blue bar) are highlighted. The implied-to-realised volatility ratio uses 3-month implied volatilities and 1-month (around 21 trading days) realised volatilities for each asset.

Asset	Current	Low	Low date	High	High date		Upside	Downside	Implied/realized volatility
S&P 500	14%	14%	03-Sep-26	24%	27-Mar-26		10%	0%	1.25x
EuroSTOXX	15%	14%	05-Jan-26	26%	27-Mar-26		12%	1%	1.08x
Nikkei 225	24%	20%	24-Sep-25	33%	31-Mar-26		9%	5%	1.28x
Hang Seng	18%	17%	04-Sep-26	24%	26-Mar-26		6%	1%	1.49x
MSCI EM	23%	14%	09-Dec-25	38%	01-Jul-26		15%	8%	1.07x
Gold	22%	16%	03-Oct-25	33%	26-Mar-26		12%	6%	1.10x
Oil (brent)	47%	27%	11-Nov-25	89%	30-Mar-26		42%	20%	1.18x
Copper	21%	17%	23-Sep-25	37%	03-Feb-26		16%	4%	0.75x
BB commodity index	18%	13%	10-Dec-25	25%	30-Mar-26		7%	5%	1.49x
EUR/USD	5%	5%	15-Jun-26	8%	13-Mar-26		2%	0%	1.63x
USD/NOX	8%	8%	17-Dec-25	11%	30-Mar-26		3%	0%	1.34x
USD/JPY	8%	7%	16-Jul-26	10%	27-Mar-26		2%	2%	0.74x
GBP/USD	6%	6%	14-Aug-26	8%	27-Mar-26		2%	0%	1.69x
USD/CHF	7%	6%	29-May-26	9%	29-Jan-26		2%	0%	1.42x
10y US swaps	81%	62%	27-Jan-26	96%	27-Mar-26		16%	18%	1.22x
10y Eur swaps	70%	43%	26-Feb-26	97%	27-Mar-26		27%	27%	1.20x
CDX IG	37%	37%	21-Sep-26	62%	27-Mar-26		26%	0%	1.96x
CDX HY	36%	31%	09-Jan-26	57%	30-Mar-26		22%	4%	1.98x
iTraxx	39%	37%	09-Jan-26	70%	27-Mar-26		31%	2%	1.51x
iTraxx X/O	40%	38%	16-Jun-26	66%	27-Mar-26		26%	2%	1.48x

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

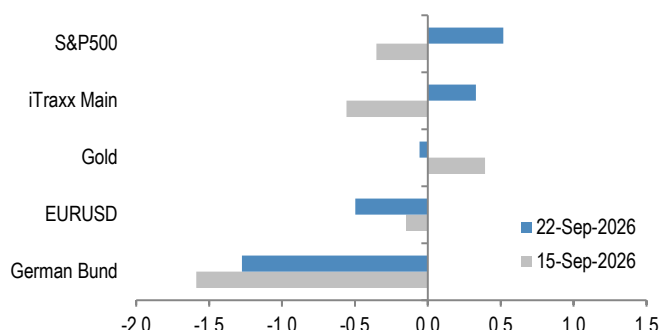
Note: Swaps volatility is 3m 10y payer ATM implied annualized BP vol and credit volatility is 3m 5y on-the-run ATM spread volatility. MSCI EM, Gold, Oil, Copper, BB Commodity Index and Treasury futures are 3m implied vol from Bloomberg.

Definitions:

Current:	Latest available closing level (21-Sep-26)
Low:	Lowest closing level in the last 1y
Low date:	Date the lowest closing level was reached (or the first time it was reached in the case of several identical low closing levels)
High:	Highest closing level in the last 1y
High date:	Date the highest closing level was reached (or the first time it was reached in the case of several identical high closing levels)
Graph:	Shows the current level and the 25th/75th percentile relative to the 1y high/low
Upside:	Implied return/volatility percentage points from current level up to the High (note: return is calculated as simple difference for spread products)
Upside (σ):	Upside in terms of standard deviations (Upside / Current 1y realized volatility)
Downside:	Implied return/volatility percentage points from current level down to the Low (note: return calculated as simple difference for spread products)
Downside (σ):	Downside in terms of standard deviations (Downside / Current 1y realized volatility)
Implied/realized volatility:	Current 3m implied volatility / current realized 3m volatility

Chart A11b: Option skew monitor

Skew is the difference between the implied volatility of out-of-the-money (OTM) call options and put options. A positive skew implies more demand for calls than puts and a negative skew, higher demand for puts than calls. It can therefore be seen as an indicator of risk perception in that a highly negative skew in equities is indicative of a bearish view. The chart shows z-score of the skew, i.e. the skew minus a rolling 2-year avg skew divided by a rolling two-year standard deviation of the skew. A negative skew on iTraxx Main means investors favour buying protection, i.e. a short risk position. A positive skew for the Bund reflects a long duration view, also a short risk position.

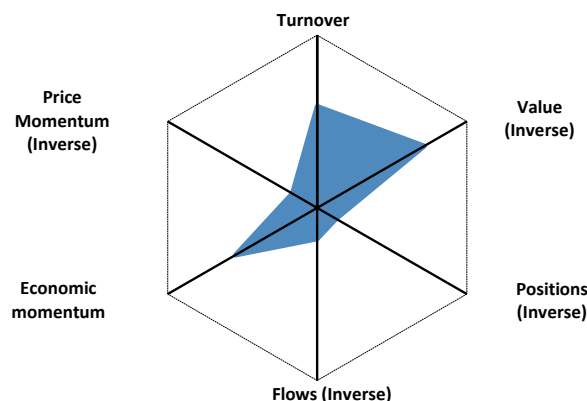


Source: J.P. Morgan Flows & Liquidity.

Chart A11c: Equity market health map

Each of the six axes corresponds to a key indicator for markets. The position of the blue line on each axis shows how far the current observation is from the extremes at either end of the scale. Overall, the larger the blue area within the hexagon, the better the outlook for the S&P500. All variables are expressed as the percentile of the distribution that the observation falls into. I.e. a reading in the middle of the axis means that the observation falls exactly at the median of all historical observations.

Turnover: Percentile rank of the 21-day rolling S&P500 turnover since 2007. **Value:** Inverse of percentile rank of our quarterly S&P 500 Equity Risk Premium (ERP) estimate since 1957 as described in "A fair value model for US Bonds, Credit and Equities", Panigirtzoglou and Loeys, Jun 2005. **Positions:** Percentile rank of the difference between government bonds and equity positioning from the cross-asset positioning monitor on the front page of our publication since 2016. **Flows:** Inverse of percentile rank of the difference between equity and bond fund flows from the cross-asset fund flow monitor on the front page of our publication since 2007. **Economic momentum:** Percentile rank of the 2-month change in the global manufacturing PMI since 1998, see "REVISITING: Using the Global PMI as trading signal", Nikolaos Panigirtzoglou, Jan 2012). **Price momentum:** Inverse of the percentile rank of the z-score of the difference in the momentum signal of S&P500 vs. 10y USTs since 2016 (based on the average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 in the Appendix). China onshore A-share ETFs have been excluded. Last obs. as of 18th Sep.

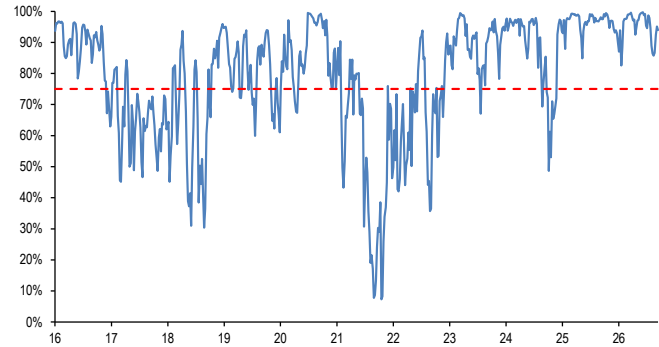


Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

24 September 2026

Chart A11d: Implied probability of the S&P500 rising over the next six months with a cutoff set at 75%

Based on the methodology laid out in the [F&L, 25th June 2025](#).



Source: J.P. Morgan Flows & Liquidity.

Chart A11e: Logistic regression model for direction of the S&P 500 over a 6 month horizon & Ex-ante probability of an up-move from the logistic regression algorithm vs. the ex-post realised 6-month return in the S&P 500

In %. Based on the methodology laid out in the [F&L, 25th June 2025](#).

Dependent Variable: S&500 UP (1) or DOWN (0) over the next six months
 Method: ML - Binary Logit (Newton-Raphson / Marquardt steps)

Sample (adjusted): 1/08/2016 12/20/2024
 Included observations: 468 after adjustments

Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
C	3.42	0.84	4.08	0.00
TURNOVER	2.97	0.72	4.15	0.00
VALUE	-3.47	0.90	-3.88	0.00
POSITIONS	-0.77	0.55	-1.39	0.16
FLows	-1.79	0.53	-3.38	0.00
ECON_MOMENTUM	3.35	0.62	5.42	0.00
MOMENTUM	-1.36	0.62	-2.19	0.03

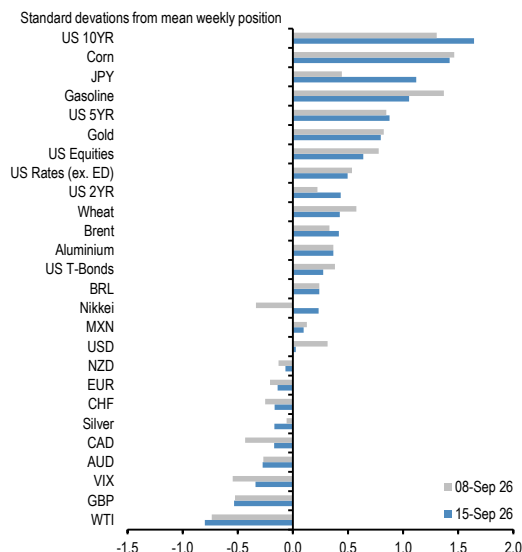


Source: J.P. Morgan Flows & Liquidity.

Spec position monitor

Chart A12: Weekly Spec Position Monitor

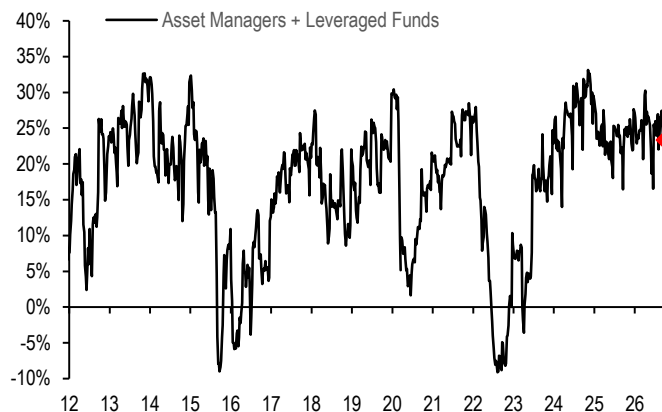
Net spec positions are proxied by the number of long contracts minus the number of short contracts using the speculative category of the Commitments of Traders reports (as reported by CFTC). To proxy for speculative investors for equity and US Treasury bond futures positions we use Asset managers and leveraged funds (see Chart A13), whereas for other assets we use the legacy Non-Commercial category. This net position is then converted to a dollar amount by multiplying by the contract size and then the corresponding futures price. We then scale the net positions by open interest. The chart shows the z-score of these net positions. US rates is a duration-weighted composite of the individual UST futures contracts excluding the Eurodollar contract.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A13: Positions in US equity futures by Asset managers and Leveraged funds

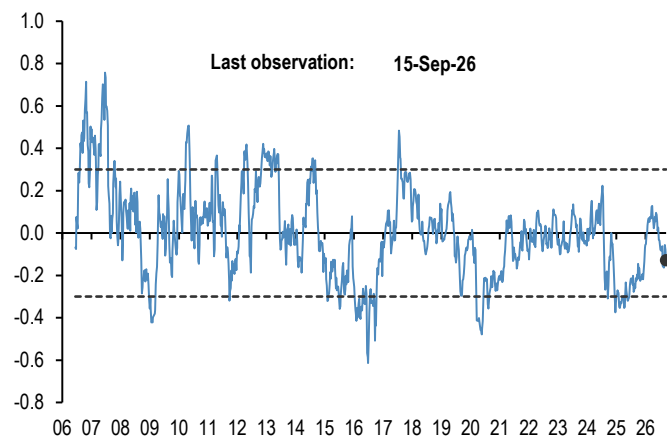
CFTC positions in US equity futures by Leveraged funds and Asset managers (as a % of open interest). It is an aggregate of the S&P500, DowJones, NASDAQ and their Mini futures contracts.



Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

Chart A14: Spec position indicator on Risky vs. Safe currencies

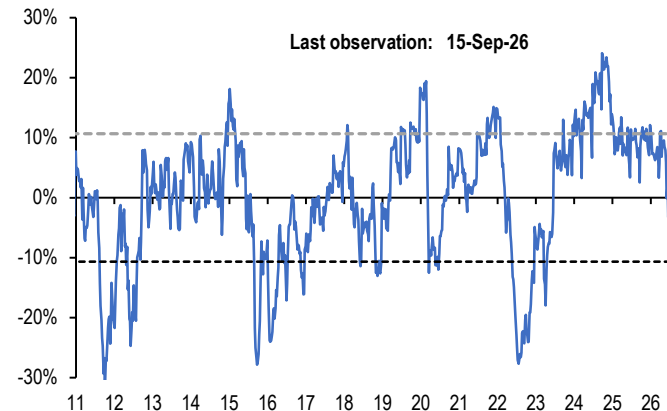
Difference between net spec positions on risky & safe currencies. Net spec position is calculated in USD across 5 'risky' and 3 'safe' currencies (safe currencies also include Gold). These positions are then scaled by open interest and we take an average of 'risky' and 'safe' assets to create two series. The chart is then simply the difference between the 'risky' and 'safe' series. The final series shown in the chart below is demeaned using data since 2006. The risky currencies are: AUD, NZD, CAD, MXN and BRL. The safe currencies are: JPY, CHF and Gold.



Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

Chart A15: Spec position indicator on US equity futures vs. intermediate sector UST futures

Difference between net spec positions on US equity futures vs. intermediate sector UST futures. This indicator is derived by the difference between total CFTC positions in US equity futures by Asset managers + Leveraged Funds scaled by open interest minus the Asset managers + Leveraged Funds spec position on intermediate sector UST futures (i.e. all UST futures duration weighted ex ED and ex 2Y UST futures) also scaled by open interest.



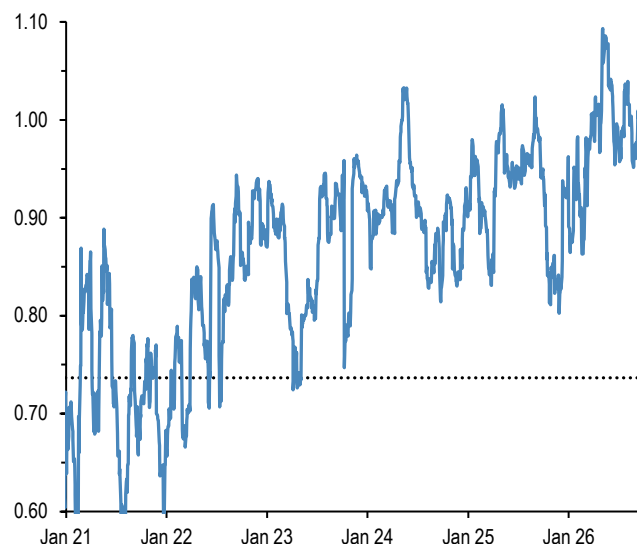
Source: CFTC, Bloomberg Finance L.P. and J.P. Morgan Flows & Liquidity.

24 September 2026

Mutual fund and hedge fund betas

Chart A16: 21-day rolling beta of 20 biggest active US bond mutual fund managers with respect to the US Agg Bond Index

The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A18: Performance of various types of investors

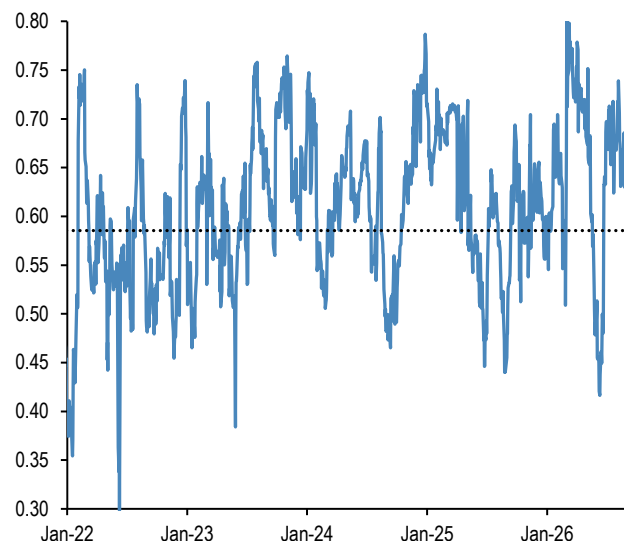
The table depicts the performance of various types of investors in % as of 22nd Sept 2026.

Date	2020	2021	2022	2023	2024	2025	2026
Investors							
Equity L/S	13.4%	9.3%	-7.0%	11.3%	13.0%	15.5%	8.3%
Macro ex-CTAs	4.9%	4.3%	11.5%	1.3%	6.0%	10.2%	6.1%
CTAs	1.2%	8.8%	14.7%	-2.5%	2.9%	-1.1%	9.4%
Risk Parity Funds	3.5%	4.7%	-18.6%	6.0%	5.0%	17.3%	8.1%
US Balanced MFs	13.2%	14.4%	-13.0%	13.8%	11.4%	14.4%	7.8%
Benchmark							
MSCI AC World	16.3%	16.4%	-18.4%	22.2%	17.5%	22.3%	14.8%
Barclays Global Agg	5.6%	-2.5%	-11.2%	7.1%	3.4%	4.9%	-0.3%
S&P Riskparity Vol 10	11.5%	12.8%	-16.2%	15.0%	5.4%	15.9%	16.9%
60 US Equity : 40 US Bonds	13.3%	14.8%	-15.4%	18.6%	16.4%	12.7%	8.4%

Source: Bloomberg Finance L.P., HFR, Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A17: 21-day rolling beta of 20 biggest active Euro bond mutual fund managers with respect to the Euro Agg Bond Index

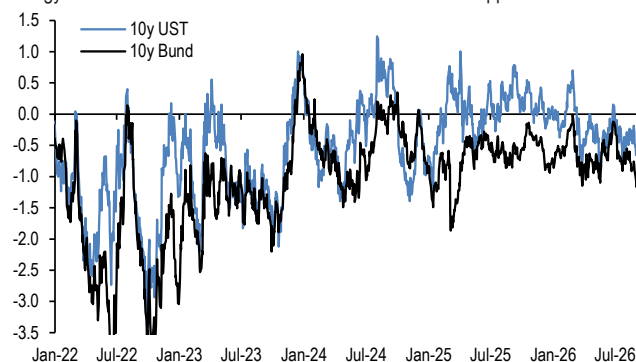
The dotted line shows the average beta since 2013.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A19: Momentum signals for 10Y UST and 10Y Bunds

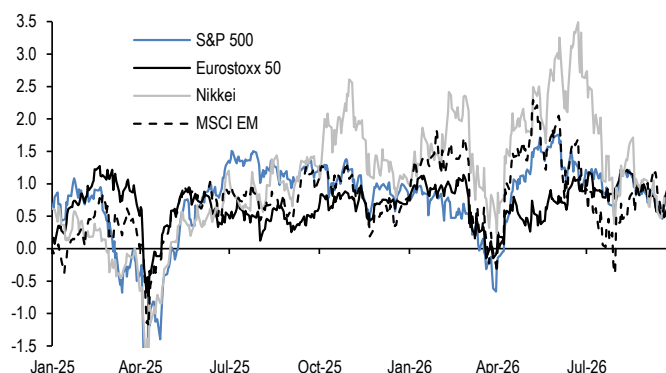
Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A20: Momentum signals for S&P500

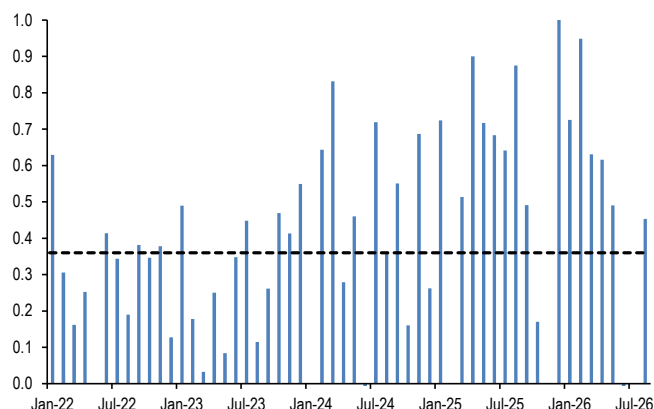
Average z-score of Short- and Long-term momentum signal in our Trend Following Strategy framework shown in Tables A3 and A4 below in the Appendix.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A22: Equity beta of monthly reporting Equity Long/Short hedge funds

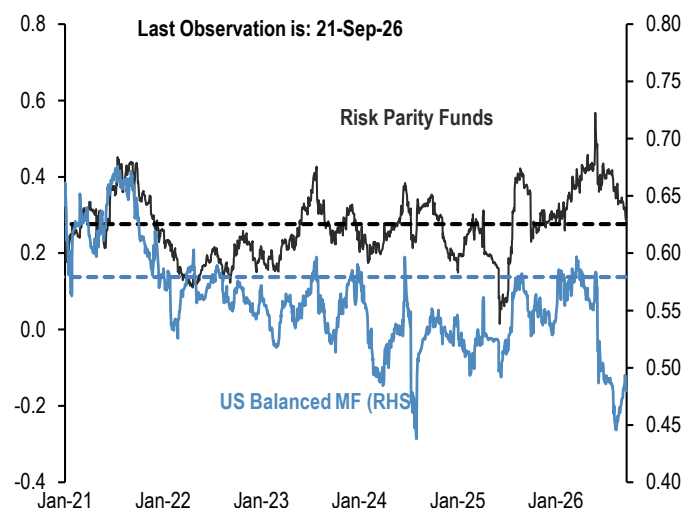
Proxied by the ratio of the monthly performance of Pivotal Path Asset-Weighted Equity Diversified hedge fund index divided by the monthly performance of MSCI AC World Index. Latest obs. is Aug 26 based on provisional data.



Source: Bloomberg Finance L.P., Pivotal Path, J.P. Morgan Flows & Liquidity.

Chart A21: Equity beta of US Balanced Mutual funds and Risk Parity funds

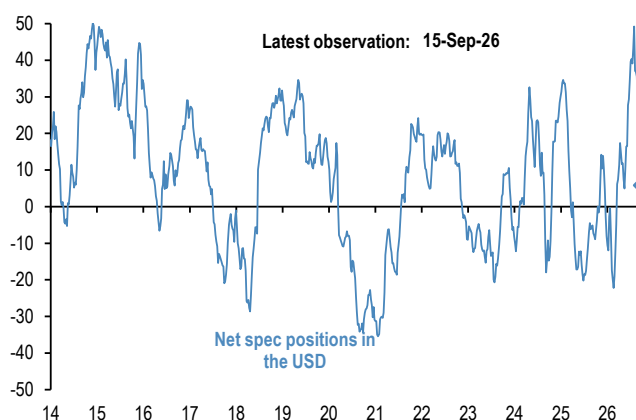
Rolling 42-day equity beta based on a bivariate regression of the daily returns of our Balanced Mutual fund and Risk Parity fund return indices to the daily returns of the S&P 500 and BarCap US Agg indices. Given that these funds invest in both equities and bonds we believe that the bivariate regression will be more suitable for these funds. Our risk parity index consists of 40 daily reporting Risk Parity funds. Our Balanced Mutual fund index includes the top 20 US-based active funds by assets and that have existed since 2006. Our Balanced Mutual fund index has a total AUM of \$700bn, which is around half of the total AUM of \$1.5tr of US based Balanced funds which we believe to be a good proxy of the overall industry. It excludes tracker funds and funds with a low tracking error. Dotted lines are average since 2015.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A23: USD exposure of currency hedge funds

The net spec position in the USD as reported by the CFTC. Spec is the non-commercial category from the CFTC.



Source: CFTC, Barclays, Datastream, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

CTAs – Trend following investors' momentum indicators

Table A3: Simple return momentum trading rules across various commodities

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

		Lookback (moving avg. days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
WTI	short	21	6.3	22.8	0.28	-1	0	-0.3	-1.7%
	long	462				0	16	1.6	46.9%
Brent	short	84	4.4	21.4	0.21	1	2	1.4	16.7%
	long	504				0	17	2.1	59.0%
Unleaded gas	short	105	3.6	23.3	0.15	0	0	1.6	21.8%
	long	483				0	62	2.9	79.9%
Heat Oil	short	63	4.5	22.0	0.21	0	18	2.0	21.1%
	long	483				0	146	3.8	119.1%
Gasoil	short	63	8.9	21.3	0.42	0	18	2.7	29.1%
	long	378				0	62	4.2	125.5%
Nat gas	short	105	13.4	35.8	0.37	0	0	0.0	0.8%
	long	315				-1	167	-0.4	-11.3%
Emission allowances	short	42	10.7	28.8	0.37	1	26	0.3	3.1%
	long	483				1	75	0.3	11.3%
TTF	short	21	43.8	43.5	1.01	-1	1	-0.2	-2.6%
	long	294				1	2	1.2	72.2%
Gold	short	21	3.2	10.7	0.30	-1	13	-0.8	-2.3%
	long	483				1	78	0.7	8.6%
Silver	short	10	6.2	19.7	0.31	0	0	0.0	0.2%
	long	462				1	69	0.8	18.1%
Palladium	short	42	11.1	23.1	0.48	-1	9	-0.4	-3.1%
	long	273				-1	91	-0.6	-14.6%
Platinum	short	105	3.1	18.7	0.17	-1	0	-0.1	-1.0%
	long	273				-1	9	-0.4	-6.3%
Aluminium	short	105	5.7	15.1	0.38	-1	67	-0.3	-2.4%
	long	357				1	72	0.6	8.8%
Copper	short	147	7.5	17.3	0.44	1	159	0.7	9.1%
	long	399				1	165	1.2	25.7%
Lead	short	126	0.8	19.2	0.04	-1	77	-0.1	-1.4%
	long	357				-1	178	-0.3	-6.7%
Nickel	short	42	11.7	22.9	0.51	-1	18	-0.2	-1.4%
	long	336				-1	9	-0.1	-2.2%
Zinc	short	126	8.3	19.6	0.42	1	127	0.9	10.3%
	long	399				1	257	1.1	26.2%
Wheat	short	168	4.0	23.3	0.17	1	14	0.7	8.8%
	long	294				1	14	0.9	14.7%
Kansas wheat	short	147	8.5	20.9	0.41	1	10	0.8	9.6%
	long	483				1	57	0.9	18.5%
Corn	short	63	7.6	16.6	0.46	1	10	1.0	7.9%
	long	399				1	22	0.3	6.0%
Soybeans	short	42	4.9	14.6	0.34	1	8	1.0	5.2%
	long	231				1	30	0.8	10.9%
Cotton	short	168	5.5	18.5	0.30	1	74	0.3	4.6%
	long	483				1	39	0.3	6.2%
Sugar	short	63	6.5	21.6	0.30	1	7	1.0	8.8%
	long	252				1	34	0.8	15.3%
Coffee	short	63	8.2	24.0	0.34	-1	11	-1.2	-10.1%
	long	273				-1	11	-0.4	-8.4%
Cocoa	short	10	5.3	31.5	0.17	-1	14	-1.7	-6.4%

* For cocoa, uses only short-term momentum and a z-score threshold of 3 rather than 1.5 as for other contracts.

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Table A4: Simple return momentum trading rules across international equity indices, bond futures and FX

Optimal look-back period of each momentum strategy combined with a mean reversion indicator that turns signal neutral when momentum z-score more than 1.5 standard deviations above or below mean, and a filter that turns neutral when the z-score is low (below 0.05 and above -0.05) to avoid excessive trading. Look-backs, current signals and z-scores are shown for shorter-term and longer-term momentum separately, along with performance of a combined signal. Annualised return, volatility and information ratio of the signal; current signal; and z-score of the current return over the relevant look-back period; data from 1999 onward.

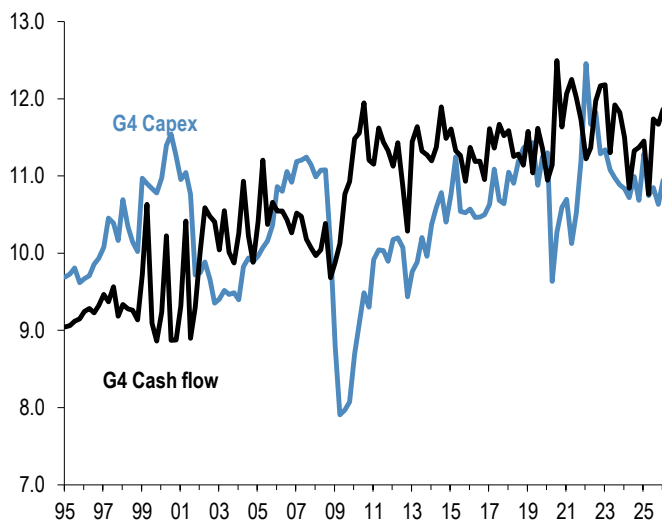
		Lookback (moving avg. days)	Annualized return (%)	Vol (%)	IR	Current signal	Time since last change (days)	Z-score	% Change of return index from its moving average
S&P 500	short	84	7.8	12.3	0.63	1	4	0.5	2.6%
	long	315				1	71	1.1	11.8%
Nasdaq 100	short	84	9.5	16.2	0.59	1	3	0.6	4.5%
	long	462				1	60	1.3	25.9%
Nikkei	short	63	3.5	13.3	0.26	-1	25	-0.3	-1.6%
	long	420				0	125	1.9	29.7%
FTSE 100	short	168	5.4	12.1	0.45	1	144	0.4	2.6%
	long	504				1	1	1.4	16.3%
Eurostoxx 50	short	168	4.8	16.6	0.29	1	120	0.4	3.9%
	long	294				1	127	0.7	7.9%
MSCI EM	short	42	12.7	11.5	1.10	1	5	0.4	2.0%
	long	357				1	52	1.2	19.2%
LQD credit return	short	168	0.4	3.4	0.12	1	2	0.1	0.3%
	long	273				1	1	0.1	0.2%
HYG credit return	short	10	2.4	6.5	0.37	1	5	0.1	0.1%
	long	189				-1	16	-0.1	-0.3%
2Y USTs	short	252	0.9	1.1	0.81	-1	147	-1.4	-1.6%
	long	462				-1	145	-1.0	-1.9%
5Y USTs	short	252	1.6	2.8	0.56	-1	140	-1.4	-3.3%
	long	420				-1	111	-1.0	-3.6%
10Y USTs	short	42	2.1	3.6	0.58	-1	20	-1.2	-1.6%
	long	441				-1	111	-0.9	-4.1%
2Y Schatz	short	189	0.6	0.8	0.81	0	28	-2.3	-1.9%
	long	504				0	144	-2.1	-3.8%
5y Bobl	short	63	1.5	1.9	0.77	0	10	-2.0	-1.9%
	long	483				0	9	-1.6	-5.4%
10y Bund	short	105	2.3	3.4	0.68	-1	2	-1.4	-3.0%
	long	462				-1	746	-1.2	-6.2%
10Y JGB	short	126	0.9	2.3	0.39	-1	3	-1.3	-1.4%
	long	273				0	146	-2.2	-3.4%
10Y Gilts	short	42	2.4	4.1	0.59	-1	5	-0.5	-0.8%
	long	399				-1	5	-1.3	-6.8%
10y OAT vs Bund	short	189	0.5	1.5	0.32	0	4	-2.3	-1.9%
	long	294				0	1	-1.7	-1.6%
10y BTP vs Bund	short	84	2.2	6.2	0.35	-1	10	-0.3	-0.6%
	long	273				-1	4	-0.1	-0.5%
Euro	short	42	3.3	6.4	0.51	-1	7	-0.8	-1.8%
	long	273				-1	155	-0.8	-4.8%
Yen	short	21	3.5	6.3	0.55	-1	2	-0.7	-1.0%
	long	462				-1	39	-1.1	-9.3%
Sterling	short	168	1.6	7.1	0.23	-1	18	-0.6	-2.5%
	long	294				-1	149	-0.6	-3.3%
AUD	short	42	3.7	7.8	0.48	-1	1	-0.3	-0.9%
	long	420				1	174	0.2	1.8%
CAD	short	168	1.5	5.9	0.25	-1	102	-0.9	-3.4%
	long	504				-1	384	-1.0	-6.6%
G10 FX	short	42	2.1	5.1	0.40	1	5	0.5	0.8%
	long	189				1	7	0.4	1.5%
EMCI	short	42	2.5	4.9	0.50	-1	1	-0.2	-0.4%
	long	189				0	0	0.0	0.2%

Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Corporate Activity

Chart A24: G4 non-financial corporate capex and cash flow as % of GDP

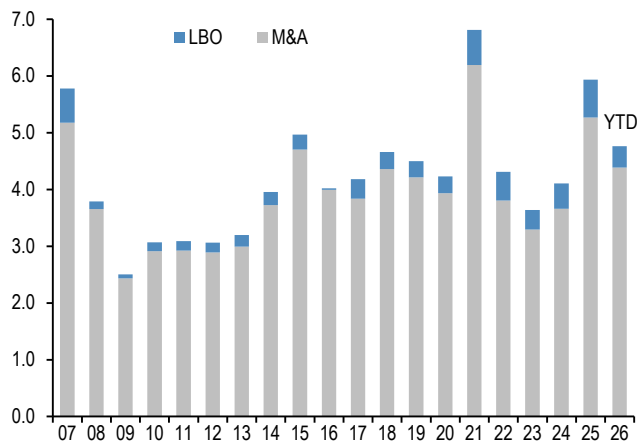
% of GDP, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q1 2026.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A26: Global M&A and LBO

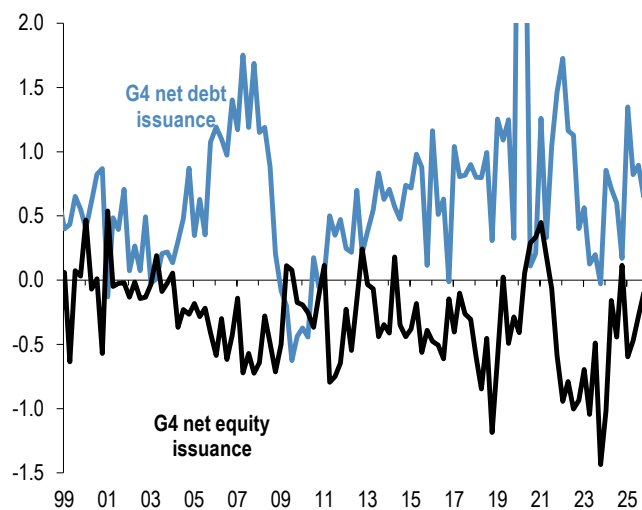
\$tr. M&A and LBOs are announced.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Chart A25: G4 non-financial corporate sector net debt and equity issuance

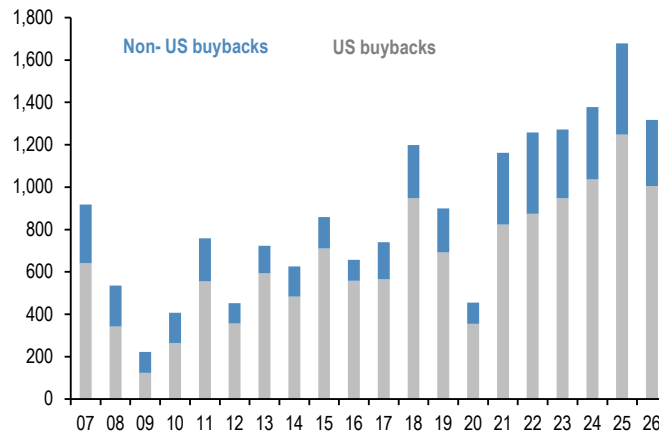
\$tr per quarter, G4 includes the US, the UK, the Euro area and Japan. Last observation as of Q1 2026.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A27: US and non-US share buyback

\$bn, are as of Aug 26. Buybacks are as announced.



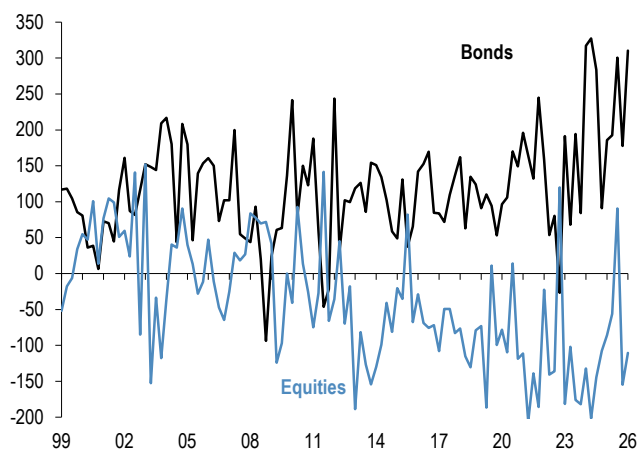
Source: Bloomberg Finance L.P., LSEG, J.P. Morgan Flows & Liquidity.

24 September 2026

Pension fund and insurance company flows

Chart A28: G4 pension funds and insurance companies equity and bond flows

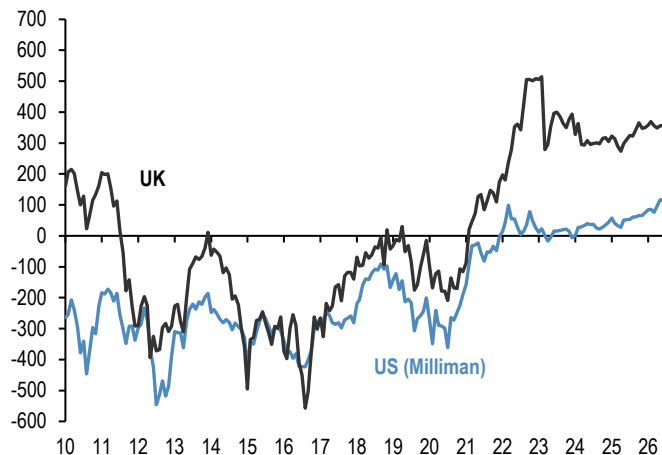
Equity and bond buying in \$bn per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q1 2026.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A30: Pension fund deficits

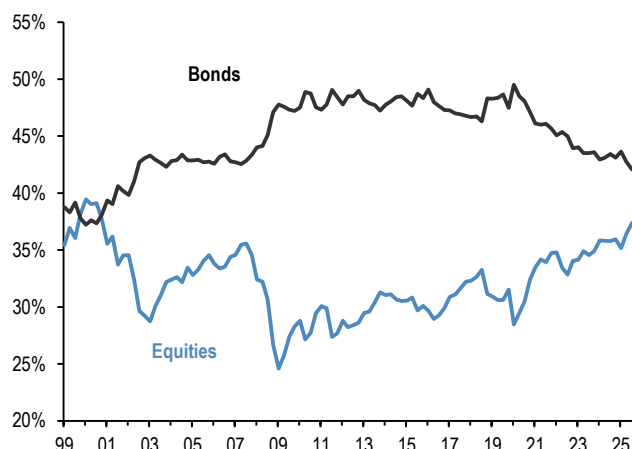
US\$bn. For US, funded status of the 100 largest corporate defined benefit pension plans, from Milliman. For UK, funded status of the defined benefit schemes eligible for entry to the Pension Protection Fund, converted to US\$ at today's exchange rates. Last obs. is July'26 for US & UK.



Source: Milliman, UK Pension Protection Fund, J.P. Morgan Flows & Liquidity.

Chart A29: G4 pension funds and insurance companies equity and bond levels

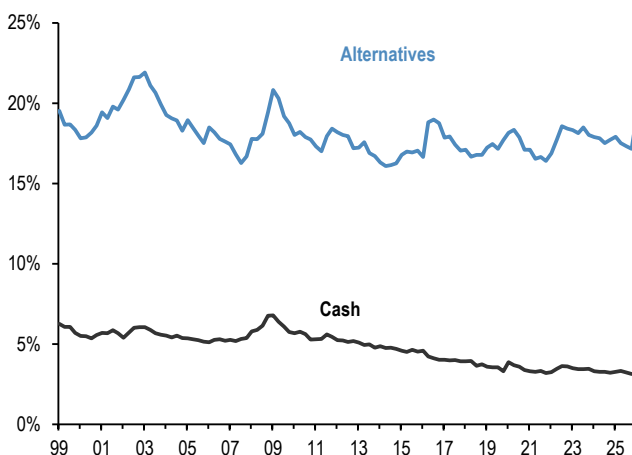
Equity and bond as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q1 2026.



Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Chart A31: G4 pension funds and insurance companies cash and alternatives levels

Cash and alternative investments as % of total assets per quarter. G4 includes the US, the UK, Euro area and Japan. Last observation is Q1 2026.

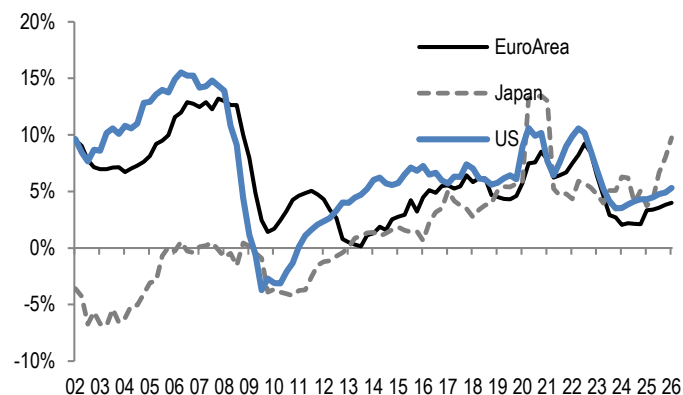


Source: ECB, BOJ, BOE, Federal Reserve flow of funds, J.P. Morgan Flows & Liquidity.

Credit Creation

Chart A32: Credit creation in the US, Japan and Euro area

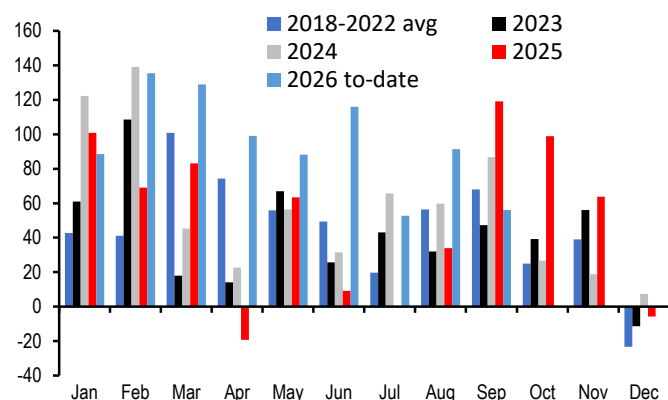
Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. Last obs. is Q1'26 for Japan, Euro Area and US.



Source: Fed, ECB, BoJ, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A34: Monthly net issuance of US HG bonds

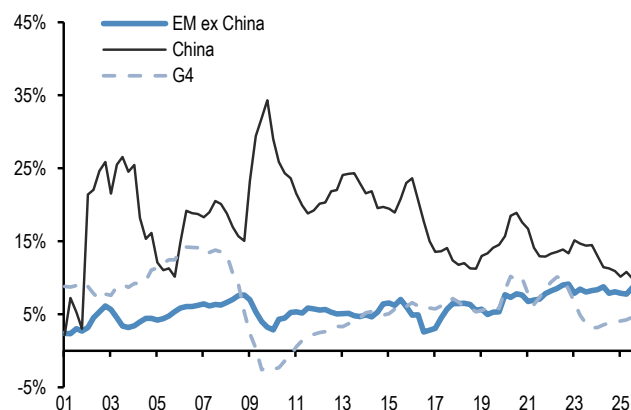
\$bn. Up to September 17th.



Source: Dealogic, J.P. Morgan Flows & Liquidity.

Chart A33: Credit creation in EM

Rolling sum of 4-quarter credit creation as % of GDP. Credit creation includes both bank loans as well as net debt issuance by non-financial corporations and households. The chart uses the FX rates as of the latest observation to calculate the credit creation at constant exchange rates. Last obs. is for Q4'25.



Source: G4 Central banks FoF, BIS, ICI, Barcap, Bloomberg Finance L.P., IMF, and J.P. Morgan Flows & Liquidity.

Table A5: Equity and Bond issuance

\$bn. Equity supply and corporate announcements are based on announced deals, not completed. M&A is announced deal value and buybacks are announced transactions. Y/Y change is change in YTD announcements over the same period last year.

Equity Supply	11-Sep	4 wk avg	13 wk avg	y/y chng
Global IPOs	1.3	2.3	4.4	162%
Secondary Offerings	12.1	10.8	15.7	44%
Corporate announcements				
M&A - Global	120.9	97.1	116.3	34%
- US Target	43.7	37.7	57.2	50%
- Non-US Target	77.2	59.4	59.1	22%
Net bond issuance	Feb-26	3 mth avg	YTD avg	y/y chng
USD	8	144	179	-50%
Non-USD	162	129	277	-4%

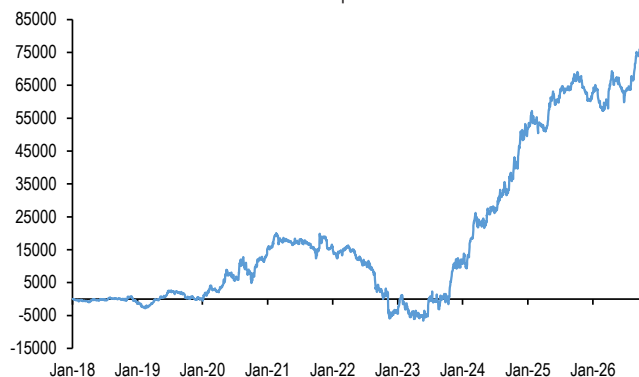
Source: Bloomberg Finance L.P., Dealogic, Thomson Reuters, J.P. Morgan Flows & Liquidity.

24 September 2026

Bitcoin monitor

Chart A35: Our Bitcoin position proxy, based on open interest in CME Bitcoin futures contracts

In number of contracts. Last obs. for 22nd September 2026.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A37: Ratio of bitcoin market price to production cost

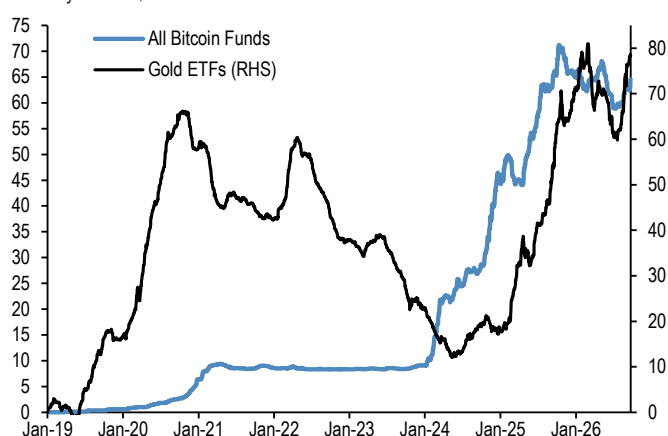
Based on the cost of production approach following Hayes (2018).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A36: Cumulative Flows in all Bitcoin funds and Gold ETF holdings

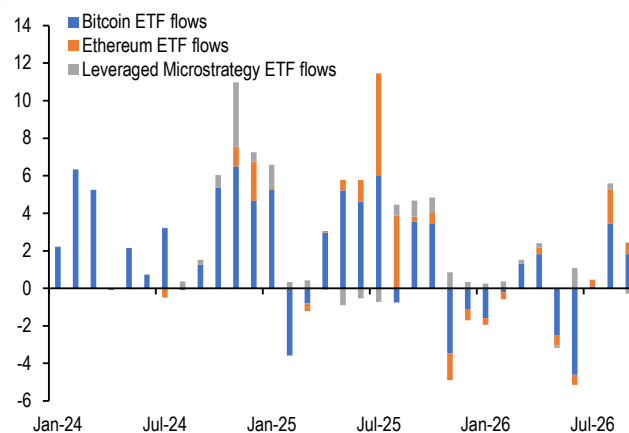
Both the y-axis in \$bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A38: Monthly flows into spot bitcoin ETFs, spot ethereum ETFs and leveraged MicroStrategy ETF

\$bn.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Japanese flows and positions

Chart A39: Tokyo Stock Exchange margin trading: total buys minus total sells

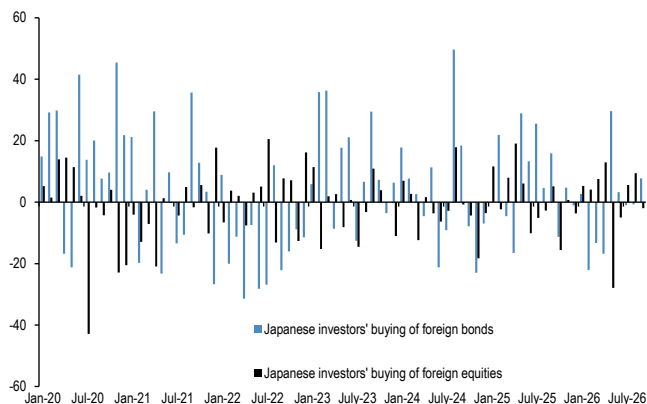
In bn of shares. Topix on right axis.



Source: Tokyo Stock Exchange, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A41: Monthly net purchases of foreign bonds and foreign equities by Japanese residents

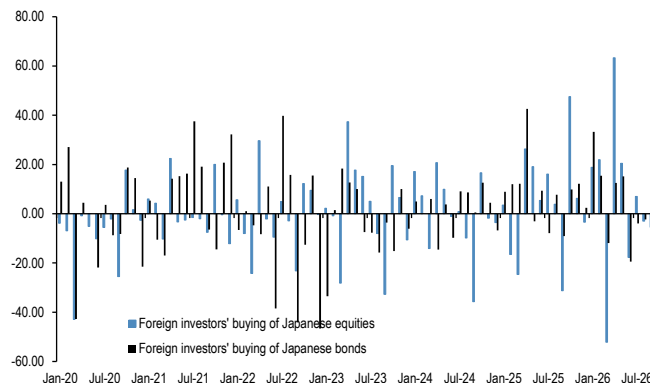
\$bn, Last weekly obs. is for 11th Sep' 26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A40: Monthly net purchases of Japanese bonds and Japanese equities by foreign residents

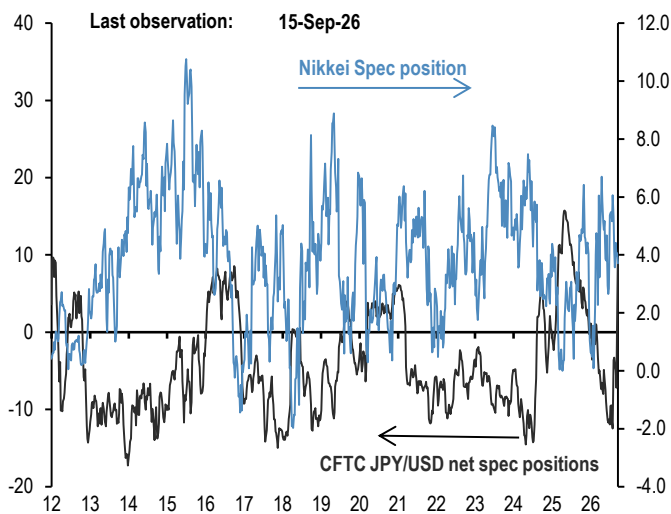
\$bn, Last weekly obs. is for 11th Sep' 26.



Source: Japan MoF, Bloomberg Finance L.P., and J.P. Morgan Flows & Liquidity.

Chart A42: Overseas CFTC spec positions

CFTC spec positions are in \$bn. For Nikkei we use CFTC positions in Nikkei futures (USD & JPY) by Leveraged funds and Asset managers.



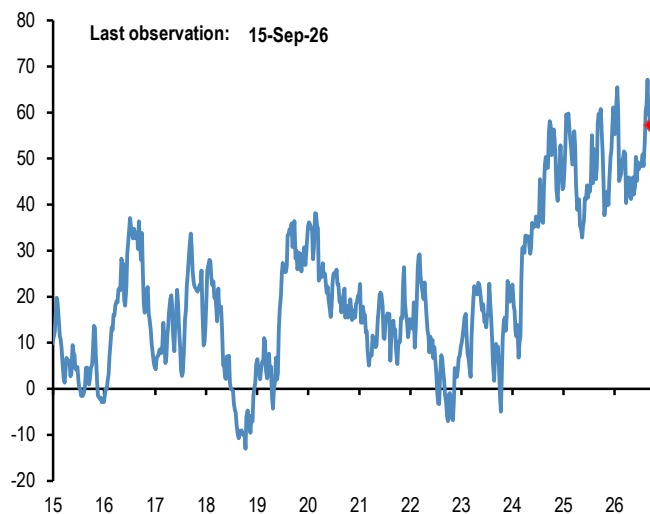
Source: Bloomberg Finance L.P., CFTC, J.P. Morgan Flows & Liquidity.

24 September 2026

Commodity flows and positions

Chart A43: Gold spec positions

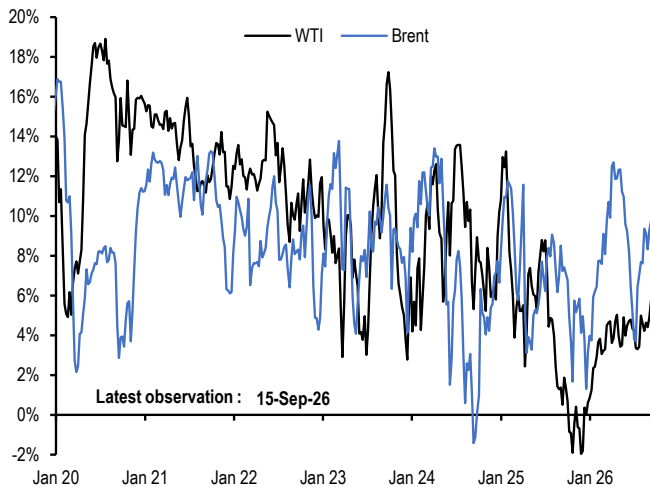
\$bn. CFTC net long minus short position in futures for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A45: Oil spec positions

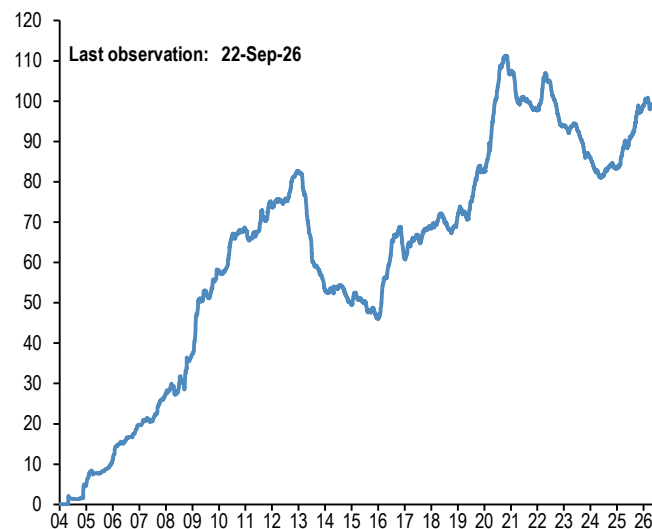
Net spec positions divided by open interest. CFTC futures positions for WTI and Brent are net long minus short for the Managed Money category.



Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A44: Gold ETFs

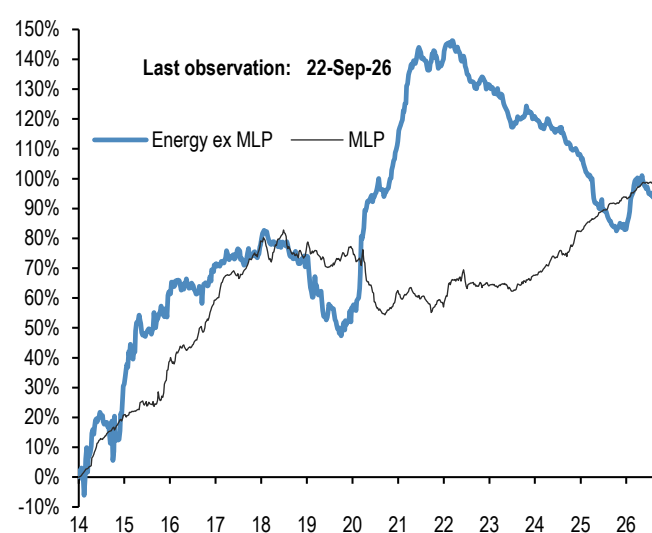
Mn troy oz. Physical gold held by all gold ETFs globally.



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A46: Energy ETF flows

Cumulative energy ETFs flows as a % of AUM. MLP refers to the Alerian MLP ETF.

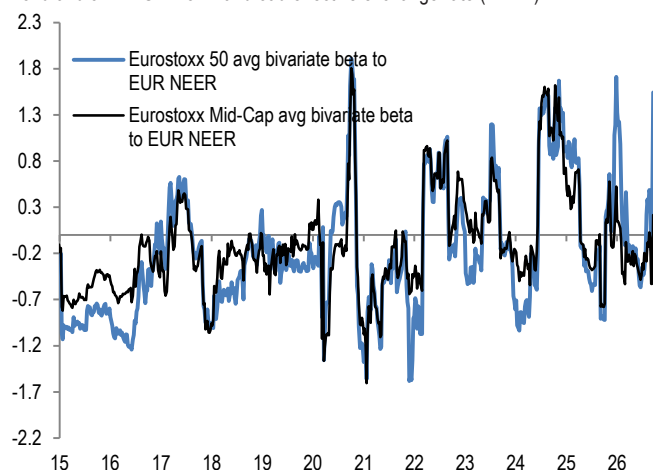


Source: CFTC, Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Corporate FX hedging proxies

Chart A47: Average beta of Eurostoxx 50 companies and Eurostoxx Mid-Cap to trade-weighted EUR

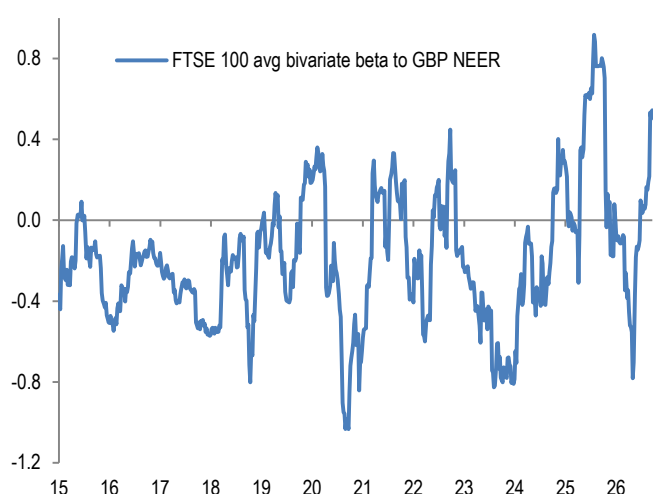
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the Eurostoxx 50 index to the weekly returns of the MSCI AC World and JPM EUR Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A49: Average beta of FTSE 100 companies to trade-weighted GBP

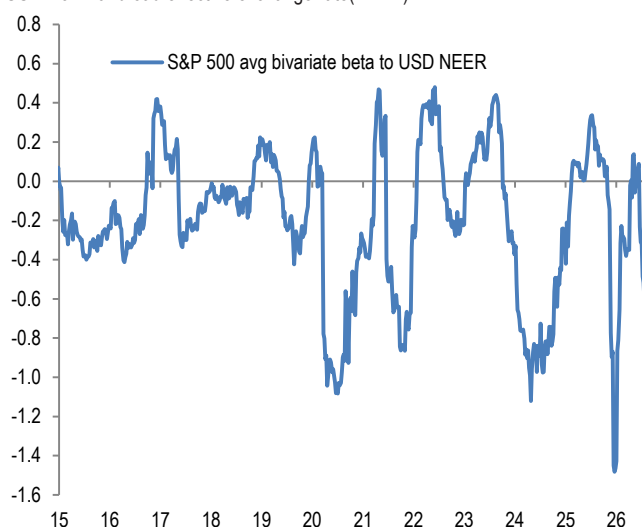
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the FTSE 100 index to the weekly returns of the MSCI AC World and JPM GBP Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A48: Average beta of S&P500 companies to trade-weighted US dollar

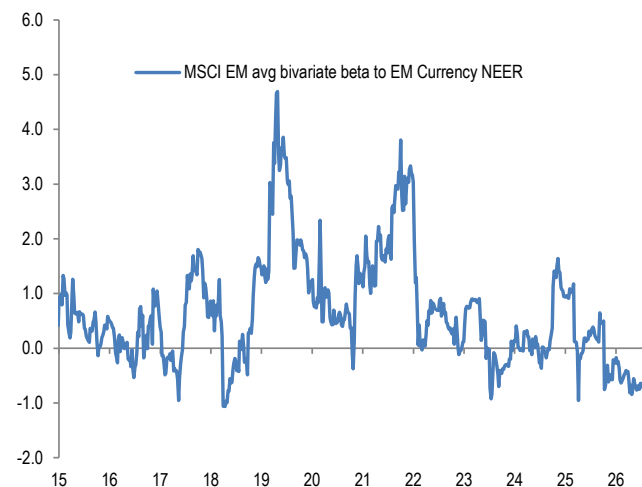
Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of stocks in the S&P500 index to the weekly returns of the MSCI AC World and JPM USD Nominal broad effective exchange rate (NEER).



Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

Chart A50: Average beta of MSCI EM companies to trade-weighted EM Currency Index

Rolling 26 weeks average betas based on a bivariate regression of the weekly returns of individual stocks in the MSCI EM index to the weekly returns of the MSCI AC World and JPM EM Nominal broad effective exchange rate (NEER).



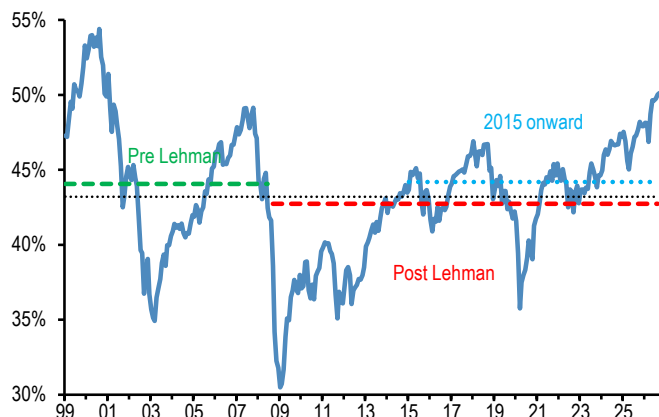
Source: Bloomberg Finance L.P., J.P. Morgan Flows & Liquidity.

24 September 2026

Non-Bank investors' implied allocations

Chart A51: Implied equity allocation by non-bank investors globally

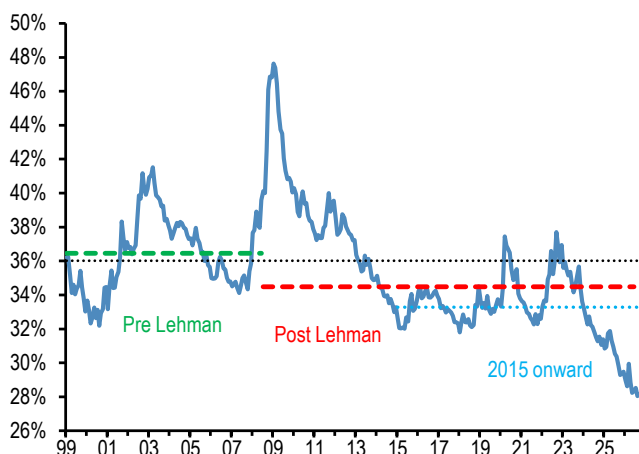
Global equities as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

Chart A53: Implied cash allocation by non-bank investors globally

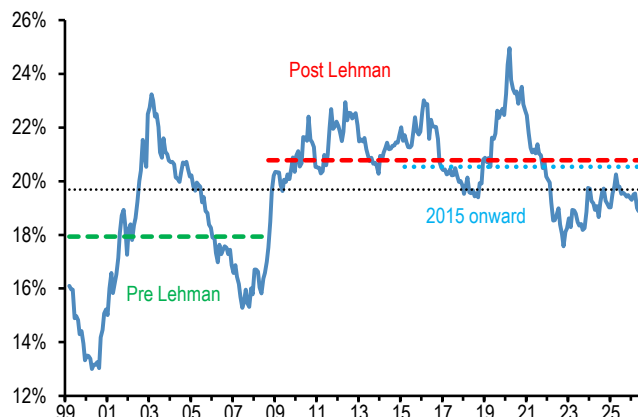
Global cash held by non-bank investors as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

Chart A52: Implied bond allocation by non-bank investors globally

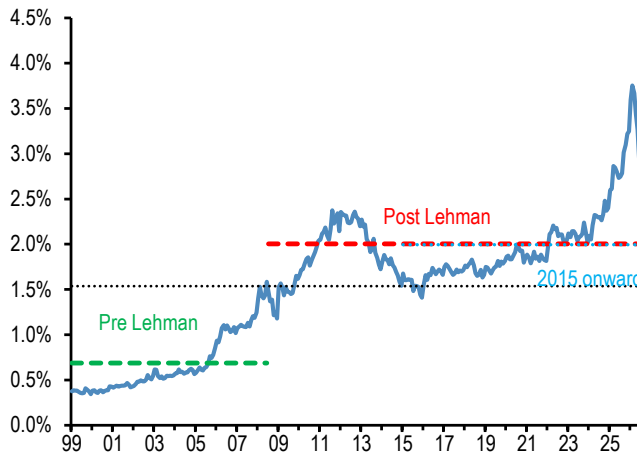
Global bonds as % total holdings of equities/bonds/M2 by non-bank investors/Commodities including gold held by private investors. Dotted lines are averages.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

Chart A54: Implied commodity allocation (including privately held gold) by non-bank investors globally

Proxied by the open interest of commodity futures ex gold and privately held gold as % of the stock of equities, bonds, cash held by non-bank investors globally and commodities including gold held by private investors.



Source: Bloomberg Finance L.P., LSEG DataStream, J.P. Morgan Flows & Liquidity.

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	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	36%	12%
IB clients**	83%	80%	73%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	92%	87%

*Please note that the percentages may not add to 100% because of rounding.

**Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.

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